

ALL PEOPLE ARE CREATIVE

A primer on generative design research

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PREFACE

What people need

This book is dedicated to the memory of Ivan Illich, the radical Austrian philosopher and critic of the institutions of contemporary Western culture, who so succinctly described, in *Tools for Conviviality*, the most basic and underserved need of people.

People need not only to obtain things, they need above all the freedom to make things among which they can live, to give shape to them according to their own tastes, and to put them to use in caring for and about others. (Illich, 1975, page 11)

Illich goes on to introduce a concept that he calls *convivial tools*. To understand what he means by convivial tools, it is necessary to consider his definition of *tools*. He uses the word ‘tools’ to refer to *all rationally designed devices* including hardware, machines, procedures, *productive institutions such as factories that produce tangible commodities like corn flakes or electric currents, and productive systems for intangible commodities such as those which produce ‘education’, ‘health’, ‘knowledge’ or ‘decisions’*. Tools are the man-made consequences of design and development processes. Illich defines convivial tools by contrasting them to industrial tools. (Illich, 1975, page 20)

Convivial tools are those which give each person who uses them the greatest opportunity to enrich the environment with the fruits of his or her vision. Industrial tools deny this possibility to those who use them and they allow their designers to determine the meaning and expectations of others. Most tools today cannot be used in a convivial fashion. (Illich, 1975, page 21)

Tools for Conviviality was first published in the United States in 1975 (and in Great Britain in 1973), at a time when Illich’s insights and implications for the future fell on deaf ears. And so now almost 40 years later, the situation in industrialized parts of the world is at least as bad as Illich imagined it would be. Our tangible and intangible commodities have been tuned and tweaked over those years with increasingly sophisticated industrial tools that promote efficiency and productivity over happiness and creativity. New technologies for automation have taken time and steps as well as worker pride out of the production processes for goods and services. The consumptive mindset has been passed on from generation to generation, growing stronger and more powerful in each transfer. This is particularly true in the United States. In part, the design profession is to blame. As designers, we have become masters at creating and disseminating what Illich describes as “*useful things for useless people*” (Illich, 1975, page 34). It is time to acknowledge that people want to be useful and creative and not just spend their time shopping, buying and consuming.

In introducing the concept of convivial tools, we are not necessarily talking about people making their own products (i.e., end-use fabrication) although that might be an example. The concept of convivial tools is much broader than that. We can interpret and envision convivial tools at many different levels of abstraction to include, for example:

- end-user fabrication of products
- platforms for the creative expression of individuals
- the means for achieving community goals
- methods, tools and techniques for exploring what conviviality might look like and feel like
- scaffolds for collective creativity

Imagine what we will learn about convivial tools over the next 50 years!

What needs to happen?

It is time to balance industrial tools with convivial tools. We need to learn how to balance consumptive behavior and lifestyles with more creative behavior and convivial lifestyles. The situation today is looking positive as there are a number of small points of light at the end of the tunnel.

- There is the growing realization that we can't continue living the way we have been living because we simply don't have the resources. The internet has helped us to see first-hand that the distribution of resources is not at all equally distributed, making the need to conserve all the more important, e.g., www.worldbank.org.
- The recession, difficult as it is for many people, is actually helping us to see more clearly what we really need. And it is helping us to realize how good it feels to give to others.
- There is a resurgence in interest in and emphasis on creativity at all levels: in learning, at work and at play (Pink, 2006; Florida, 2002).
- People have dreams for how they want to live and what is most important in life. And we now are beginning to see that people are choosing experiences over stuff (Pine and Gilmore, 1999).
- New information and communication technologies are connecting people to each other and facilitating collective thinking and doing. Social networks, in particular, bring the promise of facilitating more convivial ways of living.

We need to begin to explore what convivial tools will become. How can we make and use them? How can new technology support conviviality? And how can we evaluate the application of new technologies by new human-centered standards for convivial living? For example, can we measure whether the new technologies support people in their needs to:

- Be and feel creative?
- Make a contribution?
- Give shape to the things among which they can live?
- Put these things to use in caring for and about others?

What is the book about?

This book is about generative design research, an approach to bring the people we serve through design directly into the design process in order to ensure that we can meet their needs and dreams for the future. Generative design research gives people a language with which they can imagine and express their ideas and dreams for future experience. These ideas and dreams can, in turn, inform and inspire other stakeholders in the design and development process.

This book is about a future that provides for and supports the use of convivial tools. Until now, only experts such as designers, engineers and business people have been involved in making industrial tools for people. Such industrial tools include consumer products, computer systems, banking and retail services, etc. But the team of experts approach will not work for the making of convivial tools. Convivial tools must be made *with* the people, not only *for* the people who are being served through design. By co-creating with the people served through design, we can ensure that the tools we design are convivial.

An important disclaimer

There is one thing that we need to make clear from the beginning. The key idea on which the book is based is that **all people are creative**. By this we mean that all people have ideas and can contribute to design processes that aim to improve their lives as well as the lives of others. It is a key idea but it is also a controversial hypothesis and we are well aware that not everyone believes it to be true.

We will not spend time trying to convince you that everybody is creative. We will show how they can be creative in design. We will take their creativity as a given and describe how you can best involve people in the creative process starting in the front end of design. To the extent that you can already see or can come to believe that all people are creative, you will be able to maximize the potential of generative design thinking.

If you don't already believe that all people are creative, the best way to see this for yourself is to get hands-on learning experience. If you never come to see and believe that all people are creative, you can still use the tools, techniques and methods described here to inspire your own creative process.

Who is the book written for?

This book is for people who work (or who aim to work) in the front end of the design process to innovate in human-centered ways or to facilitate cultural transformation. It is for people who want to

change how we live and work and play for the better. It is for people who seek a better balance of industrial and convivial ways of living. The book is also for people who are trying to understand people in deep ways as might be the situation for applied social scientists.

In thinking about who the book is for, the categories don't follow the traditional academic disciplines. The categories are better described as mindsets. Let us explain. In many years of teaching and sharing the ideas about generative design research, we have seen four distinct mindsets emerge with regard to the key idea that all people are creative and can be involved in the ideation, design and development of new products and services that will affect their futures.

- Intuitives
- Learners
- Skeptics
- Converts

The **intuitives** already know that all people are creative. They don't need to be convinced that co-designing has value. They may, in fact, have been operating all along with a co-designing mindset without knowing there was a name for it. They are excited to find that this worldview is finally being given a formal description, which can help them to share their thinking with others.

Others are **learners**. They will come to understand the hows and whys of co-designing after a number of hands-on experiences. They may, through experience, come to see co-designing as their own mindset or they may choose to only pull out the tools and methods to support to their dominant worldview or to differentiate themselves in the marketplace.

The **skeptics** are those who do not believe that all people are creative. It is likely that they were rigorously trained to think of themselves as the experts in their domain. They are not open to co-designing with the people they consider to be less knowledgeable or less creative than they are. Or they may be those who have witnessed a failure of participatory processes before, such as in ineffective focus group or a commercial co-creation scam that lead nowhere.

The last category includes the **converts**. These are skeptics who, for one reason or another are put into a learning situation about co-designing, and question it the entire time, only to become extremely strong advocates and sometimes even evangelists at the end. It is impossible to distinguish the convert from the skeptic during the learning experience. The converts are a small group of people who might turn out to play a very important role in the evolution of human-centered innovation.

The distribution of intuitives, learners and skeptics varies greatly between different parts of the world, and across gender and generational lines. For example, you will find more intuitives among young women living in Scandinavia than you will find among American males over the age of 50 (who are more likely to be skeptics).

Would you consider yourself to be an intuitive, a learner or a skeptic? How you use the material presented in this book will probably be different in each case. As an intuitive, you may benefit most

from the frameworks that provide some order and guidance to the process. As a learner, you will want to carefully consider both the theory and the practice of generative design research. And even if you consider yourself to be a skeptic, you may find the tools and techniques to be useful. You might find that you are actually a convert once you have the opportunity to learn by doing.

How did we plan for the book to be used?

We have assumed that the primary users of the book will be university **students** at the advanced undergraduate and the graduate levels. And we have planned for the book to be used in a number of different academic departments such as design, psychology, marketing, engineering, business, communications, education, etc. The material that we cover is relevant to any discipline that aims to understand or improve the human condition. The book is particularly well-suited for academic programs that are exploring and offering interdisciplinary and transdisciplinary learning experiences. In that latter respect it may appeal to the **academic researcher in design**, in that it tries to bring together descriptions of state of the art methods, their underlying principles, and relation to practice, which in inter- and transdisciplinary areas is needed to connect the fields involved.

We also see this book as a preliminary text for the **research practitioner** working in industry who is looking to learn about new ways to approach the understanding of people in order to innovate at the front end of the design and development process. It can also be useful for **practicing designers** who find themselves becoming more and more interested in doing the research that can inform and inspire their own design process.

The book contains practical tips that at first sight are relevant for practice, not academic research. But with the increasing interest in and use of qualitative and situated research approaches, seeming ‘practicalities’ (such as do you conduct a session in the offices of the client, in the university, in a hired place, or in users’ homes?) are of importance, and should be included in the discourse.

And lastly, the book can also be used as a source of practical ideas and inspiration for those who are **already experienced** in the principles and practices of co-designing or human-centered innovation. For example, the frameworks for analysis and communication may offer new ways to think about these activities. The tips may come in handy during fieldwork. And the contributions from our colleagues around the world may provide inspiration for new ways of working.

Keep in mind that the book is intended to be a primer, i.e., it presents the most basic elements for understanding and conducting generative design research. It describes how this approach works and how to get started. It is not a recipe book that spells out in detail how to execute well-defined methods and procedures. For that, the state of the art is still developing too quickly, and also the methods cannot be mindlessly executed. The book presents a toolbox of generative research methods, tools and techniques that can be used to inform and inspire futures that will support community and conviviality.

The book is a convivial toolbox. It assumes that the people who will use the toolbox are creative.

How is the book structured and why?

Because the audiences that we are addressing are varied, we have organized the book to offer options in how it is used. There are four main parts.

1. Underlying principles. The first part introduces the basic components of generative design research and provides a broad theoretical background for support and grounding.
2. Getting a sense of scale. The second part consists of four cases, ranging from a very small project that was executed by a group of students over a few days to a very large, global project that involves many different team members working together over several months of time. The cases were selected to show the range of applications of generative design research, give a sense of scale and complexity, and illustrate how the many different aspects and stages discussed in part 3 fit together in practice. The two smaller cases describe real projects. The two larger cases are hypothetical cases that have been formed by combining bits and pieces of many different client-sponsored projects that the first author participated in over many years as the project manager in design research firms. Client-sponsored work is generally proprietary so we have taken the hypothetical case approach to introduce you to many different types of experiences that have actually taken place.
3. The mechanism. The third part is the how-to section that describes how to plan, gather, document, analyze and communicate the data from the generative design research process. There are no explicit recipes for successful generative design research in the how-to chapters because there are many options from which to choose. Instead of recipes, we offer frameworks for the reader to use in considering the most relevant options based on the objectives, time frame, skills, etc. at hand.
4. Samples from the field. Throughout the book are 50 contributions from a diverse range of people in the field. These provide examples and instantiations of tools, highlight specific aspects of theory or practice, or relate experience and practical tips. They also show how the techniques are used in different places and across disciplines.

Students will probably want to read all three parts in order. Research practitioners may be most interested in reading the cases, followed by the how-to section. Practicing designers will likely want to read all the contributions first to see if this is the way they would like to work in the future. Educators may want to link the principles of the first part to the tips of the later parts to further the fundamental understanding in their students. Experienced co-design researchers may find the frameworks for action that appear throughout the how-to section are the most useful in giving them new ways to expand their repertoire.

Acknowledgements

We have been very fortunate to have had interest in and support from people all over the world as we were writing the book. There are over 50 people who wrote either a one page or a two page contribution. This has added tremendously to the diversity of voices to share their experiences with innovation through co-creation with generative design research. The shorter, one-page contributions have been placed within the chapters to add to the content found there or to provide an alternative

perspective. The two-page contributions can be found at the ends of the chapters so as not to interrupt the flow of the reading. The people who wrote the contributions are our current or past colleagues, students, clients and co-workers who are now engaged in doing or using generative design research. The list of potential contributors grew as the book was being written and broadened to represent more parts of the world. And although some parts of the world are more advanced in their interest in and acceptance of generative design research, it is clear now that the phenomenon is universal.

There were six reviewers who took the time to read and critique the book in the first draft format over the winter of 2009-2010, and eight more who reviewed the second draft over the summer of 2010. We are incredibly grateful for that! We had reviewers from various parts of North America and Europe. The reviewers represented design and design research educators as well as recently graduated and current students. They work in academia as well as practice. Several of the reviewers also became contributors when they identified a hole in the draft version of the book that they were able to fill.

Short bios and contact information for all the contributors and reviewers can be found in the back of the book. To all of you, thanks again for your feedback and encouragement!

Part 1 Basics

Introduction to Part One

This book is about generative design research, an approach to bring the people we serve through design directly into the design process in order to ensure that we can meet their needs and dreams for the future. Generative design research gives people a language with which they can imagine and express their ideas and dreams for future experience. These ideas and dreams can, in turn, inform and inspire other stakeholders in the design development process.

Part One introduces the basic components of generative design research and provides a broad theoretical background to ground the discussion. Here we introduce the key idea of the book, i.e., the view that all people are able to bring creative contributions to design processes. We also describe a number of related theories and observations from social and cognitive science because the way generative tools and techniques are structured and conducted is based on an understanding of how people think, feel and act. After the theoretical grounding, we will introduce the basic building blocks of generative design research.

The chapters in Part One can be used as a stand-alone resource. For example, if you want to understand the why behind the tools and techniques of generative design research, these beginning chapters provide a framework for understanding, adapting, and further developing them. On the other hand, if you are looking primarily for how-to information, you may want to skip over Part One initially, and focus on Part Three. But you will want to return to it later since the first three chapters contain explanations underlying the practical information. The explanations in Part One will be called upon regularly in Part Three where we provide information on how to plan and execute generative design research. These explanations will help you to generalize from localized tips to broader applications as you explore further.

CHAPTER 1 Introduction

Overview of chapter one

In this first chapter we will describe generative design research by connecting it to theoretical and practical precedents, and situating it in current practice. We will position generative design research with regard to business and marketplace contexts as well as within recent developments/trends in design and design research. We will explain the emergence of the design research landscape, and the place of generative design research tools and techniques within it. We will also describe how the roles of the different stakeholders in the design development process have changed over time. We will conclude Chapter One by describing three perspectives on how co-creation and generative design research can be used.

Who is creative?

Everyone is creative. In fact, people are particularly creative with regard to experiences that they are passionate about such as living, playing, learning and working. But many people do not engage in creative activities in their everyday lives. They see creativity as something that is meant for children, not adults. Or they believe that only certain people such as artists, musicians or designers are capable of being truly creative. Since many adults in our society don't engage in creative activities regularly, they may not see themselves as being creative.

But people today want to be creative and to engage in creative ways of living. This hypothesis comes from the many years the first author has spent with people in their homes, workplaces and schools in her role as a generative design researcher. The single most consistent aspiration that people express for their own futures is the need to be and to feel creative. They communicate this whether they are at home or at school or at work. People want control over how they live, where they live and what kinds of products and services will best support them in doing so. The dream people have to engage in creative ways of living is not only pervasive, it has been gaining ground quickly in the past decade.

Why are people expressing a need for creativity now?

Why is the need for creativity so consistently expressed today? It is an antidote to consumerism. We have been living in a consumer-driven culture for some time now. Starting in the 1950's, consumerism has grown, leading to conspicuous consumption which has been growing ever since. People are seen as customers and consumers who live in the marketplace. Businesses are interested in people only in so far as they shop, purchase and then own and use the products or services that are designed, produced and sold to them. Design as a profession has traditionally been focused on serving industrial production for the marketplace. So design students have been trained to design stuff such as products and packages and communications so that they help people to consume goods and services.

Conspicuous consumption has resulted in many unsustainable products and practices. In fact, many consumers are not even aware of or are confused about the negative environmental impacts of their behavior. Consumerism has also led to a preoccupation in the business sector with innovation at all costs. In fact, even innovation is not enough. It is “radical innovation” (Verganti, 2009) that companies seek in order to stay ahead of the competition.

Fortunately, a countermovement to this pattern has recently become evident. The recession has made it abruptly and abundantly clear that continuous conspicuous consumption can no longer be maintained. And at the same time we see that many people are seeking ways to be socially and environmentally responsible. This is true for individuals as well as for large established corporations.

People will always be consumers of products, services and experiences. But what they need now is a balance between the consumptive activities and the ability to engage in creative activities. They need to be able to choose for themselves when to be consumers and when to be creators. They need opportunities to make better choices, including choices about how to live not only choices about how to spend and consume.

Business and design are changing

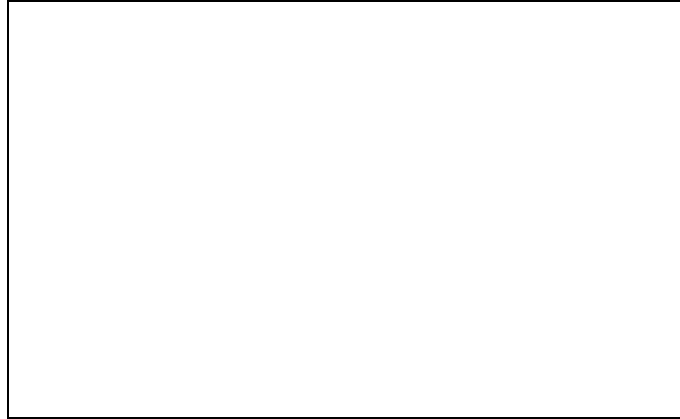
Manifestations of change can be seen at the intersection of design and business. One of these is the recent interest and enthusiasm in what is called ‘design thinking’ (Martin, 2009). Design thinking is already of such interest that business schools at universities around the world are attempting to revamp their curricula to meet the needs of business students who do not want to play the ‘business as usual’ game.

Another major sign of change is the practice of co-creation at all stages of the design development process (Sanders and Simon, 2009). Co-creation can be seen at all points along the process. The applications of the concept range from co-creation at the front end of the design process where the potential for sustainable innovation lies to co-creation at the tail end of the design development process where the potential for brand loyalty lies. We will talk more about the relationship between co-creation and its position in the design process later in this chapter.

Yet another manifestation of change at the intersection of business and design is the rise of creative activity seeking by everyday people. The growth of the DIY (do-it-yourself) industry and the resurgence of crafting at all levels (e.g., Stewart, M., 2009 and Sedaris, A., 2010) is a strong indicator that people are seeking ways to express their creativity. The rise of social networks and other means of online sharing have contributed widely to this phenomenon. www.etsy.com is a good example. The rise of creative activity seeking by everyday people may be a reaction against the overemphasis on consumption over the past 50 years. Or perhaps it is a seeking for the ‘convivial tools’ that Illich (1975) described over thirty years ago?

The changes in the business marketplace are impacting the disciplines of design and design research. The shifting foundations of the design disciplines are described in the chart below (see Figure#1.1) which shows that we are in the midst of a transformation. Design has been, until recently, primarily

concerned with the making of “stuff”. The traditional fields of design education are characterized by the type of stuff that designers learn to make (e.g., industrial designers make products, architects make buildings, etc.). Prototypes made during the traditional design process represent objects such as possible products, spaces or buildings. The languages that designers learn in school are specialized for the creating of such objects. For example, traditional design embodiments for making stuff include sketches, drawings, prototypes, and models of objects, often in isolation.



Figure#1.1: The disciplines of design are transforming from a focus on the objects of design (old) to a focus on the purpose of designing (new).

Design practice is now moving from a preoccupation with the making of stuff to a focus on making stuff for people in the context of their lives. The emerging design domains on the right side of the chart are focused on the purpose of designing, e.g., design for the purpose of serving or healing or transforming. Thus, in these new design domains, there is the need for alternative forms of conceptualization and embodiment beyond “stuff”. Some of the alternative embodiments for describing and enacting experience that are being explored today include stories, future scenarios, narratives, performance art, documentaries and timelines of experience. In addition, personas (hypothetical yet representative people) are frequently used to ensure that people are a focal point in the design process.

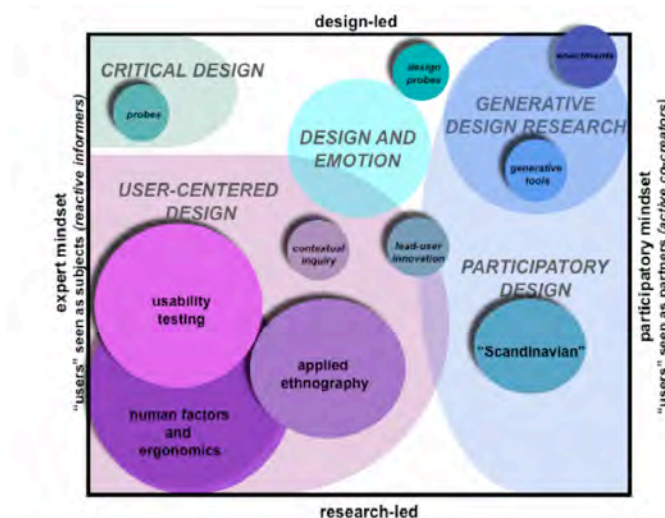
The new and emerging design domains are bigger and more ambitious than the traditional disciplines. The emerging design disciplines require the collaboration of people from many different backgrounds, including both designers and nondesigners.

The landscape of design research

In the past two decades the study of people as users of products, services and environments has grown both in industry and in practice. We will refer to this area of study as ‘design research’. In other words, it is research to inform and inspire the design and development process. The emerging

landscape of design research approaches and methods is shown in Figure#1.2 as a map. Why make a map? Making a map is a way to hold a domain still for long enough to be able to see the relationships between the various approaches, methods, and tools. Maps are good for visualizing relationships, finding dense and empty spots, and spotting opportunities. Maps can be useful for showing complexity and change. For example, the underlying landscape of the map may be relatively permanent, changing only as major forces affect it. But the tools and methods shift and change somewhat like trends. And the people who inhabit the landscape may come and go. As in the real world, some people like to stay put and others like to travel. So maps are good for layering complexity and for revealing change as it occurs.

The map of design research is defined and described by two intersecting dimensions: One is defined by approach and the other is defined by mindset. Approaches to design research have come from a research-led perspective (shown at the bottom of the map) and from a design-led perspective (shown at the top of the map). The research-led perspective has the longest history and has been driven by applied psychologists, anthropologists, sociologists, and engineers. The design-led perspective, on the other hand, has come into view more recently.



Figure#1.2: The emerging landscape of design research approaches and methods.

There are two opposing mindsets evident in the practice of design research today. The left side of the map describes a culture characterized by an expert mindset. Design researchers here are involved with designing for people. These design researchers consider themselves to be the experts, and they see and refer to people as “subjects,” “users,” “consumers,” etc. The right side of the map describes a culture characterized by a participatory mindset. Design researchers on this side work with people. They see the people as the true experts in domains of experience such as living, learning, working, etc. Design researchers who have a participatory mindset value people as co-creators in the design process and are happy to include people in the design process to the point of sharing control with

them. It can be difficult for many people to move from the left to the right side of the map (or vice versa), as this shift entails a significant cultural change.

The largest and most developed of the areas on the map is the user-centered design zone. People in this zone work to help make new product and services better meet the needs of “users.” They use research-led approaches with an expert mindset to collect, analyze, and interpret data in order to develop specifications or principles to guide or inform the design development of product and services. They also apply their tools and methods in the evaluation of concepts and prototypes. The three large areas of activity in the user-centered zone come from the applied social and behavioral sciences and/or from engineering: human factors/ergonomics, applied ethnography, and usability testing. There are also two smaller bubbles within the user-centered territory: contextual inquiry and lead-user innovation. (More information about the map can be found in Sanders, 2006).

The participatory design zone spreads across both the research-led and design-led approaches on the right side of the map. Participatory design is an approach to design that attempts to actively involve the people who are being served through design in the process to help ensure that the designed product/service meets their needs. Participatory design attempts to involve those who will become the “users” throughout the design development process to the extent that this is possible. A key characteristic of the participatory design zone is the use of physical artifacts as thinking tools throughout the process, common among the methods emanating from the research-led Scandinavian tradition.

The design and emotion bubble emerged in 1999 with the first Design and Emotion Conference in Delft, the Netherlands. It represents the coming together of research-led and design-led approaches to design research. Today Design and Emotion is a global phenomenon, with practitioners as well as academics from all over the world contributing to its development.

The critical design bubble (in the top left corner) is design-led, with the designer playing the role of the expert. The emergence of this bubble came about as a reaction to the large user-centered zone, with its pervasive focus on usability and utility. Critical design evaluates the status quo and relies on design experts to make things that provoke our understanding of the current values people hold. Critical design “makes us think” (Dunne and Raby, 2001). Cultural probes is a methodology in the critical design bubble (Gaver, Dunne and Pacenti, 1999). Probes are ambiguous stimuli that designers send to people who then respond to them, providing insights for the design process. Probes (in their original form) were intended to be a method for providing design inspiration rather than a tool to be used for understanding the experiences of others.

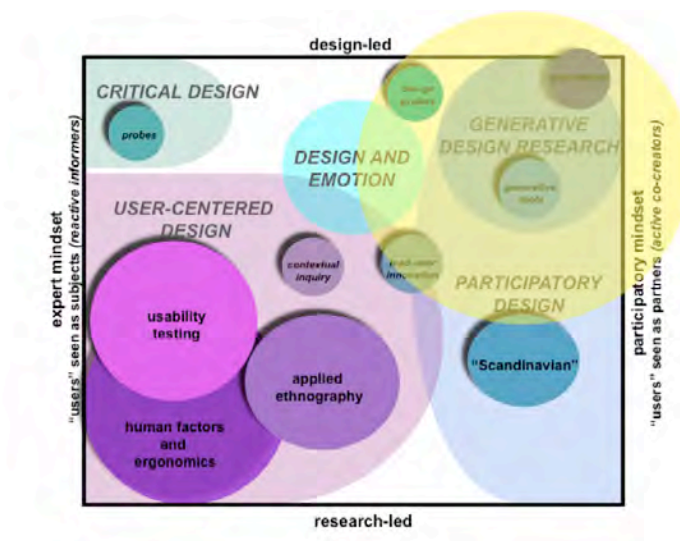
The generative design bubble (in the top right corner) is design-led and fueled by a participatory mindset. Generative design empowers everyday people to generate and promote alternatives to the current situation. The motto of this book, “all people are creative”, belongs here. The name “generative tools” refers to the creation of a shared design language that designers/researchers and other stakeholders use to communicate visually and directly with each other. The design language is generative in the sense that with it, people can express an infinite number of ideas through a limited set of stimulus items. Thus, the generative tools approach is a way to explore ideas, dreams, and insights of the people who will be served through design. Both critical design and generative design aim to generate and promote alternatives to the current situation. But they operate from opposing

mindsets. Many of the new tools and methods that have emerged in the last five years are design-led and sit along the top of the map, spanning the range from the critical design bubble to the generative design research bubble. Some examples include design probes (e.g., Mattlemaki, 2006) and various forms of enactment (e.g., Burns et al., 1996; Suri and Buchenau, 2000; Ouslasvirta et al., 2003; Simsarian, 2003; Diaz et al., 2009).

Why are the emerging design disciplines shown in Figure#1.1 not included on this map? For example, why are design for service, design for experience and design for sustainability not included as zones on the design research map? It is best to think of the emerging design disciplines as journeys that are taken across the map since they will rely on more than one location on the map. The emerging design disciplines are focused on purpose as opposed to approach or method, and so they can be addressed with either the expert or the participatory mindset. For example, the emerging discipline that is called 'service design' has adopted many of the tools, techniques and methods from participatory design and from generative design research. Time will tell as to whether service designers are practicing with a participatory mindset or whether they are using the tools and methods with an expert mindset.

Positioning this book on the design research map

In this book we will be focused on the upper right corner of the design research landscape, i.e., the generative design research territory, where the design-led approach and the participatory mindset intersect at their most extreme positions as shown in Figure#1.3. We will describe the contexts and conditions for which generative design research is most useful and relevant. We will give many examples, some from our own work and some from contributors, to make the learning experience more concrete.



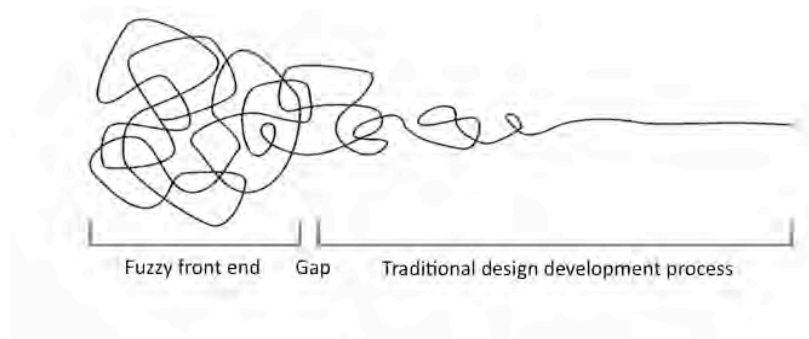
Figure#1.3: The yellow zone shows where the book is focused on the landscape of design research and practice.

We will describe the generative design research territory in relation to the other parts of the design research map but we will not describe the whole map. For that exercise we would need to write another book. We will not cover the user-centered design territory but we will talk about how preliminary exploration with traditional user-centered design methods and tools is highly recommended as a precursor to the creation and development of generative toolkits. And we will talk about how generative design research can be used to identify relevant ideas and opportunities to be further investigated and evaluated with user-centered design thinking, tools and methods.

We will not cover the entire participatory design territory in detail but we will talk about how generative design thinking and the Scandinavian approach to participatory design research are now learning from and being inspired by one another. We will not cover the emotion and design territory but we will talk about how generative tools can be used to extend the vocabulary of the people (e.g., researchers, designers and everyday people) who focus their efforts in that territory. We will also not cover the critical design territory but we will suggest potential benefits of the collaboration between these two territories in the future.

The design process is changing

The shape of the design development process has changed in response to the shifting foundations in the design and design research landscape. A very large front end has been growing and gaining in importance over the last ten years, as is shown in Figure#1.4.



Figure#1.4: The shape of the design development process has changed in response to the shifting foundations in the landscape of design research and practice.

The large front end is often referred to as the “fuzzy front end” because of its messy and chaotic nature. It is made up of the many activities that take place in order to inform and inspire the exploration of open-ended questions and to determine what are the most relevant problems to be solved. In the fuzzy front end, there is no clear path on how to proceed and there may be many divergent paths to explore before any patterns can be discerned. In the fuzzy front end it is often not

known whether the deliverable of the design process will be a product, a service, an interface, or something else. The goal of this exploration is to define the fundamental problems and opportunities and to determine what could be (or should not be) designed.

Another change in the design process is that designers are being increasingly called upon to join, and sometimes to lead, teams that are tasked with very large challenges. The types of challenges that designers are facing today are often referred to as “wicked problems”. “Wicked problems are difficult or impossible to solve because of incomplete, contradictory, and changing requirements that are often difficult to recognize. Moreover, because of complex interdependencies, the effort to solve one aspect of a wicked problem may reveal or create other problems” (Rittel and Webber, 1973).

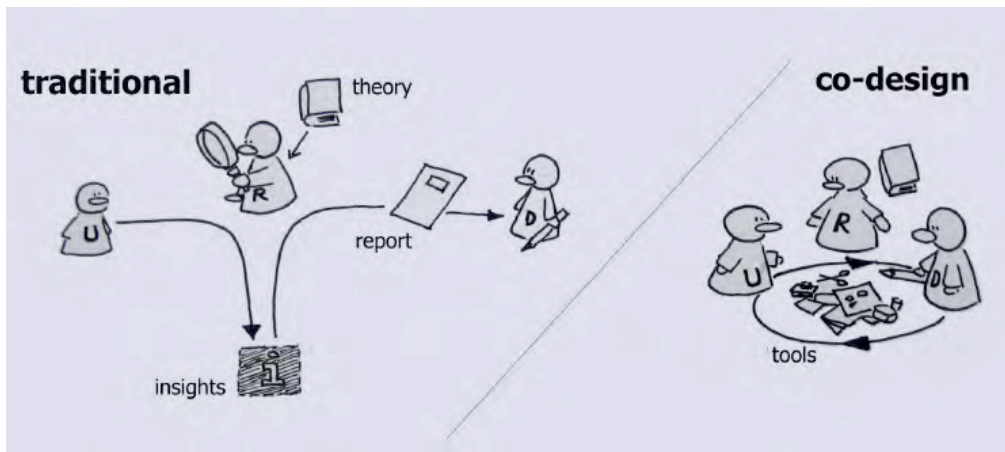
Design is not just about visualization and the application of individual creativity anymore. The problems that designers are being invited to help identify and to solve cannot be addressed by individuals, no matter how smart or creative they are. The situation is far too complex. We face significant challenges in that the problems we face are wicked and the new landscapes of design are fuzzy. But we can address the challenges of wicked problems and fuzzy pathways through collective forms of creativity and generative design thinking.

Who are the real experts when we talk about designing and innovating for future experience? We will argue that it is the people we are attempting to serve through the design process. With this shift in mindset, we can invite future “users” into the front end of the design process and move toward designing with them, not just for them. A participatory mindset can break down the disciplinary and/or cultural boundaries. Add to that the tools that can put everyone on the same playing field and support a shared language, and you have a design space that supports the exploration of new ideas, even in wicked problem situations.

People’s roles in the design process are changing

The move from the traditional user-centered design process to a design process based on collective creativity (i.e., a co-design process) is having an impact on the roles of the players in the design process. Let’s consider, for example, the roles of the user, the researcher and the designer.

Let’s start with a warning, though: there are no satisfactory definitions of these terms. Although we think we have a clear image about who we refer to with a word such as ‘user’, the literature refers to this person as ‘user’, ‘consumer’, ‘customer’, ‘insider’, ‘participant’, ‘co-creator’, ‘beneficiary’, ‘human’, ‘person’, and sometimes ‘victim’. Each word carries its nuances, is the standard term used by some disciplines, and has been criticized by others. The same holds for the roles of ‘researcher’ and ‘designer’. We don’t think that looking for the definitive term will help us much; but it helps to be alert for hidden assumptions in these words.



Figure#1.5: The roles of the researcher, designer and user change in the transition from the traditional design process to the co-design process.

The role of the user: Expert of his/her experiences

In a caricature of the traditional user-centered design process (see Figure#1.5), the user is a passive object of study, and the researcher brings knowledge from theories and develops more knowledge through observation and interviews. The designer then passively receives this knowledge in the form of a report and adds an understanding of technology and the creative thinking needed to generate ideas, concepts, etc.

In a co-design process, on the other hand, the roles change: the person who will eventually be served through the design process is given the position of ‘expert of his/her experience’, and plays a large role in knowledge development, idea generation and concept development. In generating insights, the researcher supports the “expert of his/her experience” by providing tools for ideation and expression. The designer and the researcher collaborate on the tools for ideation because design skills are very important in the development of the tools. (The designer and researcher may, in fact, be the same person). The designer continues to play a critical role in giving form to the ideas. (More on the changing roles of the key players in the co-design process is discussed in Sleeswijk Visser et al. 2005).

Users can become part of the design team as ‘expert of their experiences’ (Sleeswijk Visser et al, 2005), but in order for them to take on this role, they must be given appropriate tools for expressing themselves. Over the past decade, research groups within a number of academic institutions, practitioners in design research consultancies and design research groups within industrial institutes, have all explored co- designing tools and techniques and the processes by which they can be applied. The interest in tools and techniques for co-designing has been growing rapidly.

The role of the researcher: from translator to facilitator

In the traditional design process, the researcher served as a translator between the “users” and the designer. In co-designing, the researcher (who may be a designer) takes on the role of a facilitator. When we acknowledge that different levels of everyday creativity exist (as will be discussed in the Chapter 2), it becomes evident that we need to learn how to offer relevant experiences to facilitate people’s expressions of creativity at all levels. In addition to bringing people into the design process in the ways most conducive to their ability to participate, researchers will need to bring in applicable domain theories in ways that can be addressed by the co-design team. For example, the social psychologist who becomes a design researcher doesn’t just bring interviewing skills etc, but can also bring in background knowledge and theories about social interactions, indicating patterns and factors which can guide and/or inspire the design (e.g., Postma & Stappers, 2006).

The role of the professional designer

What role is there for professional designers to play if future users are co-creating tangible visions of new products and/or services? Some people worry that the role of the designer will become obsolete in the near future. On the contrary, design skills will become even more important in the future as the new landscapes of design emerge. A user can never fully replace a designer as designers are trained and experienced in designing. This requires specific skills. The complete process can never be outsourced.

Designers will be in demand as the usefulness of design thinking is acknowledged in our drive to address the challenges of solving wicked problems and exploring fuzzy opportunities. Designers will be needed because they hold highly developed skills that are relevant at larger levels of scope and complexity. By selection and training, most designers are good at visual thinking, conducting creative processes, finding missing information, and being able to make necessary decisions in the absence of complete information. Designers are expert prototypers and natural facilitators of collective prototyping activities.

Designers will need to play a role on co-designing teams because they provide expert knowledge that the other stakeholders don’t have. Designers professionally keep track of existing, new and emerging technologies, have an overview of production processes and business contexts. This knowledge will still be relevant throughout the design development process. Even in the design profession there is considerable specialization. The skills, knowledge and methods of the interior designer, the interaction designer, the graphic designer, etc. are quite different. These professions will not disappear overnight as ‘users’ become co-designers (Buxton, 2007). Expertise within a problem area will remain important even as new design practices emerge.

Designers will be integral to the creation and exploration of new tools and methods for generative design thinking. The onus is on designers to explore the potential of generative tools and to bring the languages of co-designing into their practice. Designers in the future will make the tools for non-designers to use to express themselves creatively.

Collective creativity and co-creation

In review, we have described how the design process is changing, particularly in the front end where the challenges of solving wicked problems and exploring fuzzy opportunities are growing, making the necessity of collective creativity more and more important. We have also discussed how people's roles in the design process are changing, with the people formerly known as “end-users” becoming acknowledged as co-creators and participants in the process because they are being recognized as the experts of their own experience.

Before moving ahead, we will define how we are using the words co-creation and co-design because you will see that the usage of these terms varies widely. We take co-creation to refer to any act of collective creativity, e.g., creativity that is shared by two or more people. Co-creation is a very broad term with applications ranging from the physical to the metaphysical and from the material to the spiritual, as can be seen by the output of search engines. By co-design we indicate collective creativity as it is applied across the whole span of a design development process. Thus, co-design is a specific instance of co-creation. Co-design refers, for some people, to the collective creativity of collaborating designers. However, we use co-design in a broader sense to refer to the creativity of designers and people not trained in design working together in the design development process. Generative design research is an approach for co-designing and co-creating that is focused on the front end of the design development process.

How and where is co-creation taking place?

The word ‘co-creation’ has been adopted across many different situations and it is often used in different ways, leading to a lack of clarity over what co-creation really is. For example, we can see:

- co-creation within communities
- co-creation inside companies and organizations often referred to as “internal co-creation”
- co-creation between companies and their business partners or other companies
- co-creation between companies and the people they serve, who are variously called customers, consumers, users or end-users

It is important to clarify up front where co-creation is taking place in order to avoid confusion in discussions about what works and what does not work.

The business community is talking a lot these days about ‘co-creation’ and the discussion here is usually framed by the ‘value’ that accrues from co-creation. It is the monetary value of co-creation that usually receives the most attention in business circles. But there are at least three types of value in co-creative activities and relationships: monetary, use/experience and societal. Co-creation that results in monetary value is fueled by the desire to make money in new ways, more efficient ways, or in ways that provide revenues over longer periods of time. Co-creation associated with monetary value may not require direct face-to-face contact between the company and its customers, for example, because the conversation can be mediated by new tools of information and communication. This can be seen in web-based surveys that ask consumers to select features of

choice or the crowd-sourcing of large numbers of respondents to obtain ideas for new products or services.

The use/experience value of co-creation is fueled by companies' desires to transform consumers into users so that the products and services they design, produce and sell will better meet people's wants and needs. One could argue that this is directly related to monetary value, but in this case the value extends beyond that of a mere monetary gain. The experience value of co-creation applies not only to products and services, but also to brands and branded environments.

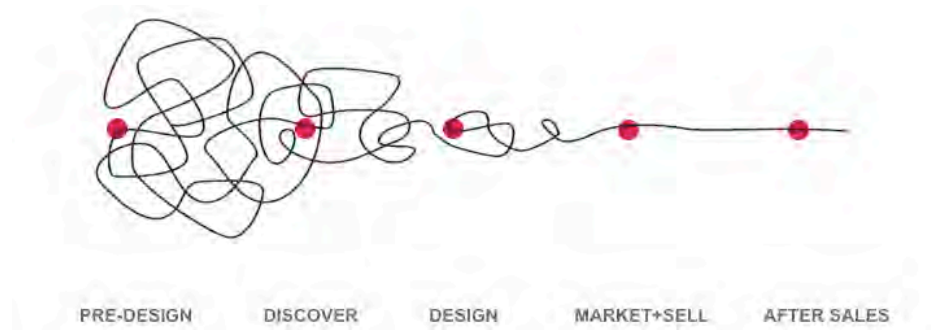
The societal value of co-creation is fueled by aspirations for longer-term and more sustainable ways of living. It supports the exploration of open-ended questions such as "how can we improve the quality of life for people living with a chronic illness?" When working within this context one does not generally have preconceived notions of the outcome since determination of the form of the outcome is part of the challenge. Thus, the problem is likely to be a wicked one. Co-creation of this type usually involves the integration of experts and everyday people working closely together. Empathy between the co-creators is essential and so face-to-face communication works best. Although social networks can be used to help identify and locate the participants, the real work in this form of co-creation favors face-to-face interactions.

All three types of value in co-creation are important to understand and to develop, and are at times inextricably linked. In fact, they can work together. Societal value can provide use/experience value as well as financial reward in the long run.

In addition to the various purposes toward which co-creation can be applied, co-creation can also take place at any stage of the design development process. (See Figure#1.6). A quick tour of Google hits to the phrase 'where is co-creation?' reveals applications of co-creation at every stage of the design process as well as many applications of the concept in the spiritual realms.

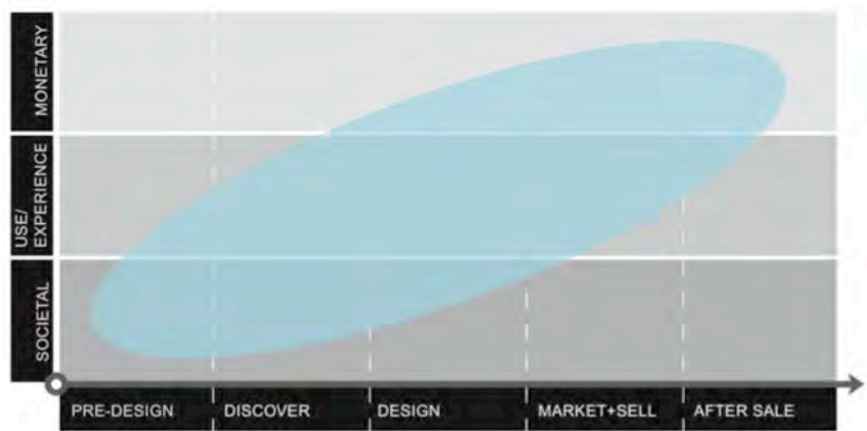
The stages of the design development process include:

- pre-design: where research, problem definition and the highest opportunity for innovation takes place
- discovery: where opportunity identification and translation of the research to design occurs
- design: where continued exploration, design, and development take place
- making: where production, building and/or manufacturing take place
- marketing, sales and/or distribution: where implementation, roll-out and sales occur
- after sales: where product use and service experience happen



Figure#1.6: Co-creation (shown by the red dots) can take place at any point along the design development process.

An interesting pattern emerges when the types of value co-creation (i.e., monetary, use/experience, societal) are aligned with the different stages of the design process. Value co-creation with a focus on monetary objectives is more likely to take place later in the design development process, such as in marketing, sales and distribution. Value co-creation of the use/experience variety tends to take place during the design exploration and design processes. And societal value co-creation is most likely to occur in the very early front end of the design process, in design exploration as shown in Figure#1.7.



Figure#1.7: There is a relationship between the level of value in co-creation and the position along the design development process.

In fact, the earlier in the design development process that co-creation occurs, the greater and broader the likely impact. Societal value co-creation tends to start at the very early front end, well before any specific concept definition and/or exploration. Practicing co-creation in the fuzzy front end will

most likely produce the largest benefit in terms of societal value. Although use/experience value and monetary value may follow, it may not become apparent until later on.

Again, in order to avoid confusion, it is important to clarify at the beginnings of conversations for what purpose co-creation is being used and where in the design development process this is happening. So to clarify, the focus of this book is on the use/experience and societal levels of value in co-creation as practiced toward the front end of the design development process.

Related approaches in co-creative design

Co-creative approaches to solving complex problems and identifying future opportunities are not without precedents, and did not emerge from any single direction. In fact, a number of related approaches are being practiced today. In this section we introduce some of these to show how similar approaches are being used by design academics, in design practice, in international development, and the social sciences. Pointers to additional resources about each of the precedents can be found in the contribution on page#X.

Participatory design

The practice of collective creativity in design has been around for nearly 40 years, going under the name ‘participatory design’. Much of the activity in participatory design has been going on in Europe. Research projects on user participation in systems development date back to the 1970s. In Norway, Sweden and Denmark, the Collective Resource Approach was established to increase the value of industrial production by engaging workers in the development of new systems for the workplace. The approach combines the expertise of the systems designers/researchers and the situated expertise of the people whose work was to be impacted by the change. The approach, thus, built on the workers’ own experiences and provided them with the resources to be able to act in their current situation (Bødker, 1996).

Participatory design practices have developed significantly since the early pioneering work. Participatory design today spans across a wide spectrum of domains and makes use of a broad repertoire of methods, tools and techniques in both commercial, community oriented and research contexts.

Positive Deviance and Lead User Design

Both in movements for social change as in design, approaches have arisen that build on the principle that among the multitude of end-users, there are some who are more capable than others of using their first-hand knowledge for creating solutions. The task of the ‘designer’ in these approaches is to channel the contributions of these participants into the larger process. Positive Deviance started in nutrition, health and emancipation work in the 1970s (Pascale, R.T., Sternin, J. and Sternin, M., 2010) The lead user approach has been developed and advocated by von Hippel (2005) and has been applied successfully in software design and other technical domains.

Action Research

From the side of the social sciences, the paradigm of Action Research has been growing since the 1940s (Susman and Evered, 1978). As the name implies, it is framed within research, and involves researchers and practitioners actively exploring possible improvements to the practitioners' situation together. A key element of the approach is an iterative sequence of interventions in work practice, and learning from this by critical reflection (Avison et al, 1999). The interventions serve two purposes simultaneously: to improve the work situation, and to understand that situation better (Gilmore et al., 1986). Action Research provides a theoretical basis for the research methodology of much academic work in this area, which is stepping outside the laboratories into practice, and benefits from its participatory and holistic viewpoints.

Contextmapping

Contextmapping started in design education in the early 2000s to bring recent developments in understanding the contexts of user-product interaction into mainstream design education. It is a procedure for conducting contextual research with users, where tacit knowledge is gained about the context of use of products (Sleeswijk Visser et al, 2005). It aims to inform and inspire design teams, where users and stakeholders actively participate in the design process to ensure a good fit between the design and the use of a future product or service. The name 'contextmapping' illustrates two main elements about the information or understanding that the design team needs: the *context* of product use, defined as 'all factors which influence the interaction between user and product'. Generative tools, as described in this book, are used for gathering user insights. The word '*mapping*' was chosen to indicate the form of this information: a tool to help access the terrain of experience, which can take many forms depending on the needs of the traveler for whom the map is made (Stappers & Sleeswijk Visser, 2006). This emphasis on communication was based on the situation in large companies and design agencies, where the design team often does not meet the participating users, but needs to be informed through techniques such as personas, scenarios, and documentaries.

Rapid and Participatory Rural Appraisal

In the area of international development, it has been seen that improvements in the life conditions of rural populations could not be imported by benevolent outsiders, but needed active involvement of the rural insiders (Chambers, 1994). Similar to participatory design, the approaches here focus on the emancipation of the role of the end-users, and a shift from informing 'outsiders' (i.e., designers) to giving control to 'insiders' (i.e., end-users).

Experience Design and Service Design

The last few years have shown a number of new design approaches that look beyond the product as a physical thing. In experience design, the focus lies on creating meaning or affect in the user, as opposed to creating a physical artifact that fulfills a functional purpose. In service design, the emphasis is on creating complex systems that create value for people through both material and organizational means. In these emerging approaches, longer-term interactions between providers and customers are taken into account, and a thorough focus on the user, and methods of co-creation are widely advocated.

The list of related approaches described in this section shows two things. First, that co-creative approaches have come up from many areas in science and design, and that the same ideas have emerged from seemingly different domains. Second, that these approaches have until now developed quite separately within the fields involved. Typically, the publications about one of the fields make little or no reference to the others, and the jargon in which the subject matter is described is also different. But in each of the areas, there is a recognition that people such as ‘end-users’ hold expertise about their needs and dreams and that their contributions are essential for finding and implementing solutions to problems. There is also a recognition of the need for appropriate, and often new, methods for facilitating them in this process.

Clearly, with the dialogue between these methods coming out in the open (also through the findability of all the above terms in Wikipedia, for instance), we can expect this field of co-creative design to mature rapidly in the coming years.

Co-creation: Is it a mindset, method or tool?

As we have discussed, the word ‘co-creation’ is being used today to describe an incredibly wide array of activities with many different goals. In fact, there appears to be somewhat of a gold rush in regard to applications of co-creation as can be seen in the increasing interest in it in the popular press. Is co-creation a new way to differentiate your company in the marketplace? Is it yet another method with an interesting collection of tools that can be called upon in the design process? Or is co-creation much larger than that? Is it a mindset (established set of attitudes held by someone), or a worldview (a philosophy of life or conception of the world) that changes how the entire design development process is seen and takes place? Co-creation can be any one or all three of these perspectives, depending on how you view it and use it.

Co-creation as a mindset. This is the broadest and most long-range of the three perspectives and the one that has the most potential to have a positive impact on the lives of people. Co-creation practiced from a mindset perspective is best executed by either very experienced generative design research practitioners or by young, intuitive practitioners. It is most useful and effective toward the front end of the design development process.

Co-creation as a method. Here we see co-creation as a collection of tools and techniques that are often compared to other collections of methods (e.g., contextual inquiry or ethnographic fieldwork). The choice of methods may depend upon who is leading the project, what the budget and timing are, and other constraints. We see co-creation as a method being used mainly during the design exploration and design phases.

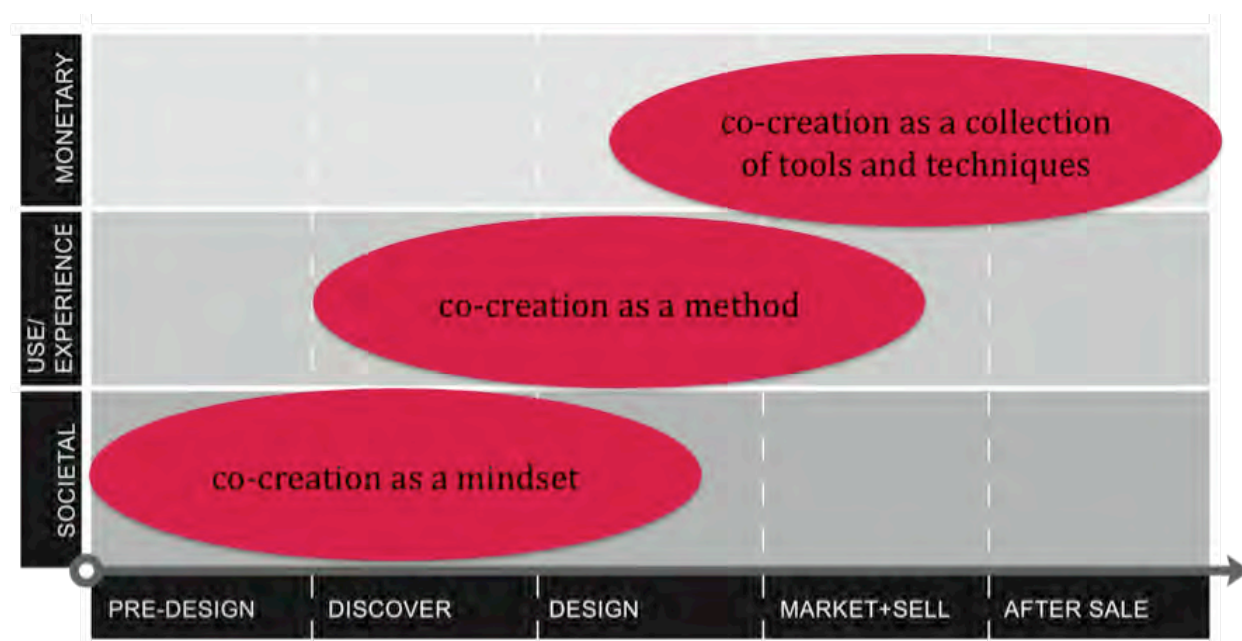
Co-creation as a tool or technique. This perspective describes the use of co-creation as just another option in the toolbox of all tools and techniques that can be used in the processes of design, development, marketing and/or distribution. For example, you often see participatory design listed on resumes of design researchers in a context similar to the one shown below.

*“My design research skills include: usability tests, personas, concept tests, task analysis, diary studies, contextual inquiry, **participatory design**, paper prototype evaluation, focus groups,*

cognitive walkthroughs, card sorting, eye-tracking, remote usability and surveys, ethnographic field studies and phone interviews.”

It is apparent from the context that the author of this list understands co-creation/participatory design to be a tool or technique that can be applied in specific situations. They are not viewing it as a mindset. Co-creation as a tool or technique is the perspective that has received the most attention in the popular press as it is being used as a fast and low-cost way to drive interest in and attention to brands and/or new products and services in the marketplace.

Figure#1.8 shows the relationship between the three perspectives on co-creation and the point in the design development process at which the perspective is most useful.



Figure#1.8: The three perspectives on co-creation are positioned by level of value co-creation and along the design development process.

Co-creation as a mindset is most useful in the front end. Co-creation as a method is useful in the design-focused phases of the process. Co-creation as a tool or technique is most useful at the tail end of the process. This is not to say that the three perspectives must be used only at those parts of the process, but that these are very good starting points.

Where this book is focused

In summary, because co-creation is such a broadly used term, it can be confusing when people use the word ‘co-creation’ from different perspectives. It is important to be clear up front about:

- who is involved in the co-creation (e.g., communities, companies, organizations, partners, people, etc.)?
- what perspective is leading the effort (e.g., co-creation as a mindset, method, tool or technique)?
- what is the value (e.g., monetary, use/experience, societal) that is of interest in co-creation?
- where in the design development process (e.g., front, middle or end) will the effort be taking place?
- how are the responsibilities distributed across all the players? For example, who can give suggestions, make interpretations, come up with ideas, etc.?

The focus in this book will be toward the front end of the design development process (to include pre-design, design exploration and design), with the co-creation as a mindset perspective and with an interest in all levels of value co-creation. The cases and contributions will describe a variety of examples showing:

- co-creation within communities,
- co-creation inside companies and organizations
- co-creation between companies and their business partners
- co-creation between companies and the people they serve, who are variously called customers, consumers, users or end-users

In the next chapters we present some fundamental psychological mechanisms underpinning creativity in general, and co-creation in particular. In the chapter after that, we show how these mechanisms find their form in generative tools and techniques.

CHAPTER 1 bubble further readings

Further readings on related approaches in co-creative design

We have touched briefly on some the main precedents for co-creative approaches and generative design research. If you are interested in reading more, you can start with the resources listed here.

Participatory Design

Ehn, P. (1988). *Work-oriented design of computer artifacts*. Falköping, Sweden: Arbetslivcentrum/Almqvist and Wiksell International.

Greenbaum, J., and Kyng, M. (1991). *Design at work: Cooperative design of computer systems*. Hillsdale NJ: Erlbaum.

Muller, M.J. (2007) Participatory design: The third space in HCI (revised). In J. Jacko and A. Sears (eds.), *Handbook of HCI* 2nd Edition. Mahway NJ USA: Erlbaum.

Sanoff, H. (2000) *Participatory design: Theory and techniques*. Bookmasters, Raleigh, NC.

Schuler, D., and Namioka, A. (1993) (Eds.). *Participatory design: Principles and practices*. Hillsdale NJ USA: Erlbaum.

Positive Deviance and Lead User Design

Pascale, R.T., Sternin, J. and Sternin, M. (2010) *The Power of Positive Deviance: How Unlikely Innovators Solve the World's Toughest Problems*, Harvard Business School Publishing.

See also <http://www.positivedeviance.org/>

Von Hippel, E. (2005), *Democratizing Innovation*, Cambridge, MA: MIT Press.

Action Research

Greenwood, D. J. & Levin, M., (1998) *Introduction to action research: social research for social change*, Thousand Oaks, California: Sage Publications.

Reason, P. & Bradbury, H., (Ed.) *The SAGE Handbook of Action Research*. Participative Inquiry and Practice. 1st Edition. London: Sage, 2001.

Reason & Bradbury, *Handbook of Action Research*, 2nd Edition. London: Sage, 2007.

Stringer, E.T. (1999). *Action research*. Thousand Oaks, CA: Sage Publications.

Contextmapping

Designing For, With and From User Experiences (2010), *Proceedings of the May 13, 2009 Symposium*. Faculty of Industrial Design Engineering, Delft University of Technology, StudioLab Press.

Sleeswijk Visser, F, Stappers, PJ, van der Lugt, R, Sanders, EB-N (2005) 'Contextmapping: Experiences from Practice'. *CoDesign*, 1(2), 119-149.

See also <http://studiolab.io.tudelft.nl/contextmapping/>

Rapid and Participatory Rural Appraisal

Chambers, R. (1994) The origins and practice of participatory rural appraisal. *World Development*, 22(7), 953-989.

Experience Design and Service Design

Norman, D. (2002) *The design of everyday things*. Basic Books

Stickdorn, M. & Schneider, J. (Eds) (2010) *This is service design thinking*. BIS Publishers, Amsterdam

Note: End of bubble.

CHAPTER 2 Thinking about creativity

2.0 Overview of chapter two

In the second chapter we will describe what you need to know about creativity to understand how generative design research works. It is possible to practice generative design research without having this level of understanding, but without it you will not be able to grow in your abilities as a generative design researcher. So if you choose not to read this chapter now, you will want to return to it after you have had one or two hands-on experiences.

You can think of this chapter as the grounding for Chapter Three (How it Works: Generative Tools and Techniques) as well as for Part Three of the book. The models, theories and principles that we present in Chapter Three apply to all parts of the generative design research process including what happens in planning the fieldwork, during generative sessions or interviews, in the analysis sessions, and during communication events.

We will start by describing what we mean when we talk about *everyday creativity*. We will discuss what creativity is all about, starting with the traditional perspective that creativity is something that happens “in the head.” Then we will introduce larger and larger perspectives on creativity as the chapter progresses. We will end with a model of collective creativity that describes how people are creative together.

2.1 What is creativity?

There is no generally accepted definition of creativity. In fact, the definition of creativity has been debated for years. There are many theories about what creativity is, how it works and who does or does not have it. But there is not a lot of agreement. A quite general definition in the area of problem solving is that an idea is regarded as creative if it is both new and an appropriate solution to the problem. But design is not only about problem solving. The Wikipedia entry on ‘creativity’ (viewed January 2011) gives good overview of this lack of clarity.

On the other hand, there is agreement on the fact that creativity is being increasingly recognized and discussed. A few of the books that have really helped spark interest in the rise of creativity include:
~ Richard Florida’s (2002) book: *The Rise of the Creative Class: And How It’s Transforming Work, Leisure and Everyday Life*. Florida uses economic data to show that regions with high concentrations of creative professionals tend have high levels of economic development. He then goes on to describe what can be done to attract the Creative Class.

~ Daniel Pink’s (2005) book: *A Whole New Mind: Why Right Brainers will Rule the World*. Pink describes the six “senses” that one should harvest to build a balanced mind: design, story, symphony, empathy, play and meaning.

~ Roger Martin’s (2009) book: *The Design of Business: Why Design Thinking is the Next Competitive Advantage*. Martin describes why organizations need to embrace design thinking in order to balance their overreliance on analytical thinking.

We will center our discussion with a much older but still much cited theory about creativity. According to Arthur Koestler (1964), every creative act involves bisociation, a process that brings together and combines previously unrelated ideas. In *The Act of Creation* (1964) he writes: *The logical pattern of the creative process is the same in all three cases; it consists in the discovery of hidden similarities. But the emotional climate is different in the three panels: the comic simile has a touch of aggressiveness; the scientist's reasoning by analogy is emotionally detached, i.e. neutral; the poetic image is sympathetic or admiring, inspired by a positive kind of emotion.*" Koestler's definition of creativity is broad enough to cover all types of creative acts, whether in art, science or humor.

Everyday creativity

Many people think of creativity as a manifestation of genius that is possessed by rare individuals and often in the arts or the sciences. Within design, and especially in problem solving, creativity is often defined as *the ability to produce novel and appropriate works* (Sternberg and Lubart, 1995). But there are other types of creativity.

Margaret Boden (2004) makes a distinction between H-creativity and P-creativity. H-Creative refers to historically creative. For example this would be someone who came up with an idea, concept or product that no one has ever thought of before. P-creative is Psychologically Creative, where someone borrows an idea from one domain and applies it to another. This type of creativity is not so unique, and it applies to everyone. Boden claims that people who have had creative ideas in the past are more likely to have them in the future and people who have new ideas frequently will continue to do so.

We will refer to P-creativity as 'everyday creativity'. A framework for everyday creativity has been developed based on many years of experience in observing and talking with people about their needs and dreams for living (Sanders, 2005). For everyday creative activities we can distinguish four levels, as shown in Figure#2.1.

	level	motivated by	purpose	example
1	doing	productivity	"getting something done"	organizing my herbs and spices
2	adapting	appropriation	"making things my own" or "make it fit better"	embellishing a ready-made meal
3	making	asserting my ability or skill	"make with my own hands"	cooking with a recipe
4	creating	curiosity	"express my ability"	dreaming up a new dish

Figure#2.1: Four levels of everyday creativity are observed when talking to people about their needs and dreams for living.

The most basic level of creativity is **doing**. The motivation behind doing is to accomplish something through productive activity. For example, people have told us that they feel creative when they are productively engaged in everyday activities such as exercising or organizing their closets. Doing requires a minimal amount of interest. The skill requirements are low as well. Many of the goods

and services offered to ‘consumers’ today can be said to satisfy the doing level of creativity. They come to the consumer readymade. For example, in the food preparation domain, a doing activity would be to buy or select a prepackaged microwave entrée and prepare it for a meal. Another cooking related example could be organizing one’s herbs and spices.

The next level of creativity, **adapting**, is more advanced. The motivation behind adapting is to make something one’s own by changing it in some way. People might do this to personalize an object so that it better fits their personality. Or they might adapt a product so that it better fits their functional needs. We can see adaptive creativity emerging whenever products, services, or environments don’t exactly fit people’s needs. Adapting requires more interest and a higher skill level than doing. It takes some confidence to go ‘outside of the box.’ Following our example in food preparation, an adapting activity might be to add an extra ingredient to a cake mix to make it special.

The third level of creativity is **making**. The motivation behind making is to use one’s hands and mind to make or build something that did not exist before. There is usually some kind of guidance involved, e.g., a pattern, a recipe, or notes that describe what types of materials to use and how to put them together. Making requires a genuine interest in the domain as well as experience. People are likely to spend a lot of their time, energy, and money on their favorite making activities. Many hobbies fit in this level of creativity. A food-related example might be to create an entrée using a recipe.

The most advanced level of creativity is **creating**. The motivation behind creating is to express oneself creatively or to innovate. Truly creative efforts are fueled by passion and guided by a high level of experience. Creating differs from making in that creating relies on the use of raw materials and the absence of a predetermined pattern. For example, making is cooking with a recipe, whereas creating is making up the dish as you go.

All people are capable of reaching higher levels of creativity, but they need the passion and the experience to do so. Consequently, people differ in the level of creativity they attain in different domains. In fact, they may find themselves at all four levels of creativity simultaneously in different domains. They may attain the higher levels of creativity only in their hobbies or other domains for which they have high interest and/or passion.

The path from doing to adapting to making and finally to creating develops in the individual over time and through experience. Many people never reach the final level of creating as they are satisfied with the creativity they express at the making level, for example. In fact, people don’t always progress through the levels in the same order. Although most of us will start with doing, some will continue to creating by way of first adapting, and then making. Others will get there by first making, then adapting. For example, in the domain of wardrobe and clothing, it is as though the act of making one’s own clothes allows the maker to see new possibilities for clothing or outfits that they already own (Meyer, 2010). Others may leave out either adapting or making, but it is rare that people move straight from doing to creating.

It is important to offer relevant experiences to facilitate people’s expressions of creativity at all levels. It takes different kinds of support at the different levels of creativity. For example, it is best

to

- lead people who are on the doing level of creativity,
- guide those who are at the adapting level,
- provide scaffolds that support and serve peoples' need for creative expression at the making level, and
- offer a clean slate for those capable of creating things from scratch.

What this means is that it will take more time and preparation for people at the doing and making levels of creativity to be able to express themselves creatively in that domain. We will introduce the path of expression framework later in the chapter to describe how you can prepare people to express themselves creatively.

Here are four principles for facilitating everyday creativity. They have proven to be very useful in establishing a common ground for the team that is embarking on generative design research. In this chapter we will provide background information (in the form of theories, frameworks and models) to support these principles.

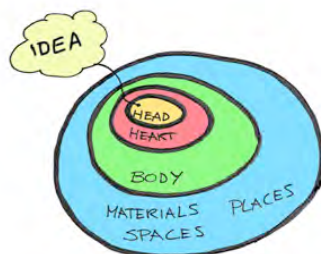
1. All people are creative.
2. All people have dreams.
3. People will fill in what is unseen and unsaid based on their own experience and imagination.
4. People project their needs onto ambiguous stimuli because they are driven to make meaning.

It will be helpful for you to reflect on these principles and discuss them with others you are collaborating with in conducting generative design research.

2.2 A framework for individual creativity

Most of the research that has been published about creativity has been about individual creativity and the leading work has been done by cognitive psychologists. So most of the theories about creativity describe what happens in the mind of one person. But this is changing. It is recognized now that creativity is not just in the head. But we will start there to keep it simple.

Figure#2.2 shows a framework for individual creativity that reveals the many layers of context around the 'in the head' perspective. It shows that individual creativity is not only in the head but in the heart as well: it involves emotion. And creativity takes place in the body. It is evoked through activity and motion. There is a timeframe for creativity and creativity can be greatly enhanced through preparation. And the last layer shows that creativity is in the environment as it is influenced by the places and spaces and through the props and materials that are available for use.



Figure#2.2: A framework for individual creativity

Generative design thinking acknowledges and makes use of all these layers of the creative context with a special emphasis on the environmental level in the form of tools and techniques and materials.

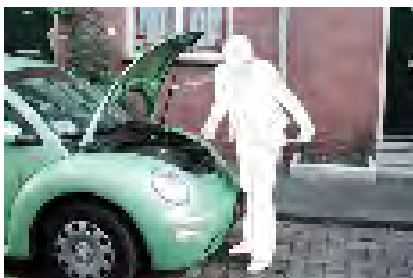
This framework is not another theory about creativity. It is a model for holding together the bits and pieces of many other theories and principles, particularly those that we felt were relevant to the workings of generative design research. We will describe later in the chapter how this framework for individual creativity expands into a model for collective creativity. But first we will discuss it layer by layer, starting with the more traditional view of creativity being something that happens in the mind of one person and describe some of what is known about ideas.

2.2.1 Ideas: The basic building blocks of creativity



Figure#2.3: At the core of the framework for individual creativity is the idea in the head.

We will start with the core, i.e., the innermost layer in the framework for individual creativity. This layer is focused on ideas in a person's head. Ideas are the basic building blocks of creativity. An idea can be represented by a word or words or it can be depicted in a picture. Take a look at the photograph in Figure#2.4 below.



Figure#2.4: There are many different ideas in this photograph of Anneke Keller and her car.

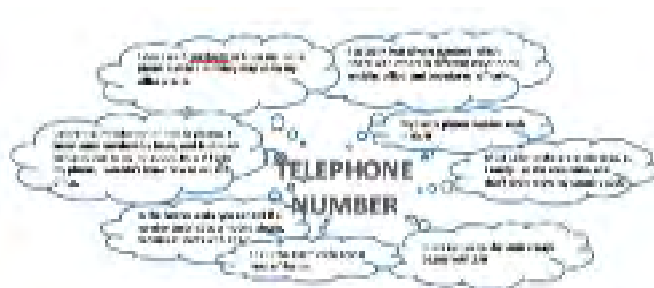
It shows Anneke Keller making a phone call, while reading the oil gauge of her car. You cannot be sure from the picture why she is making the call. She may be calling a colleague to postpone a meeting because her car won't run. Or she might be calling the garage to ask what she can do about the oil. If we consider the situation depicted in Figure#2.4, many different kinds of ideas come to

mind. For example, “I see a woman standing beside a car making a phone call”, ‘woman’, ‘car’, ‘a green Volkswagen’, ‘standing’, ‘a phone number’. Other ideas may be more difficult to express, e.g., “the freedom given by being able to call” or “how the scene makes me think of something I did last week”.

We’ll try to keep the discussion focused here, and concentrate on the simpler examples when we talk about ideas. There are basically two ways of thinking about what we mean by an idea such as ‘a phone number’.

Ideas can be defined as they are in the dictionary. The classical way to think about what an idea is is to have a strict rule such as definitions we find in a dictionary. For example, “A phone number is a 9 or 10-digit sequence, consisting of a country code, an area code, and a local number.” An example is ‘+31152785202’. It defines a telephone connection, either fixed or mobile. This definition helps us to see that ‘+31152785202’ and ‘+31152783029’ are both phone numbers, that ‘abcdefghijkl’ is not a phone number, although it consists of 10 characters. The good thing about ideas as defined by a dictionary is that there is an explicit definition, which can be the same for everybody, and which can be relied upon not to change. The bad thing is that it doesn’t explain a lot about how people deal with ideas.

Ideas can also be thought of as clouds of associations. Another way of thinking about ideas is that people carry clouds of associations that are connected to every idea they know. Some of these are close to the dictionary definition, but most of these associations have to do with their personal histories and how they have used or thought about the idea before. For example, when I think about the idea of a ‘phone number’, some of these could be:



I don't want everybody to know my home phone number, but they may know my office phone.

I actually have four phone numbers which I share with others in different ways: home, mobile, office, and secretaries' office.

My home phone number ends with 4578.



Figure#2.5: Every idea is connected to clouds of associations that are related to the idea.

The clouds of associations in Figure#2.5 would be somewhat different for every individual. These clouds of associations can be larger or smaller, and can easily be triggered. Once one cloud is remembered or triggered, nearby clouds will be remembered as well. This phenomenon is known as priming or as “spreading activation in semantic memory” (Collins and Loftus, 1975). When people are asked to quickly write down as many associations as possible to a given word or ideas, some people will quickly jot down a long list, others stop after a few words. With a bit of practice, the lists get longer due to the spreading activation of the associations in the clouds.

When we speak about a ‘phone number’, what we really mean is most often connected to these clouds of associations, rather than to the dictionary-style definition. For example, if somebody asks you: “Can I have your phone number?” he or she doesn’t just want a 10-digit sequence, but may implicitly be asking also “may I call you?”

Think of ambiguity as a tool. Ideas possess ambiguity. For example, many words are ambiguous in that they have more than one meaning. The sentence “I’ll phone you to tell you the number of phone numbers on my phone” means roughly the same as “I’ll call you to tell you how many contacts I have on my mobile” and uses three different forms of ‘phone’, and two different forms of ‘number’. But unless it gets complicated, we usually have no problem in recognizing what was intended. This means that if we present people with a single word, they may react to a variety of the meanings that are carried with the idea that the word triggers. The context they are in is likely to change how they think of the idea. For example, “time” at work may not mean the same thing as “time” at home.

Ambiguity also holds for pictures. A picture of a clock, such as the one shown here, can bring out different meanings to people depending on their past experiences and the context they are in:

- “a clock, i.e., a time-indicating device”
- “a time: ten to two”
- “something is due to start”
- “a deadline”

- “time pressure”
- “keeping track of time”
- “I forgot to go to the bank”

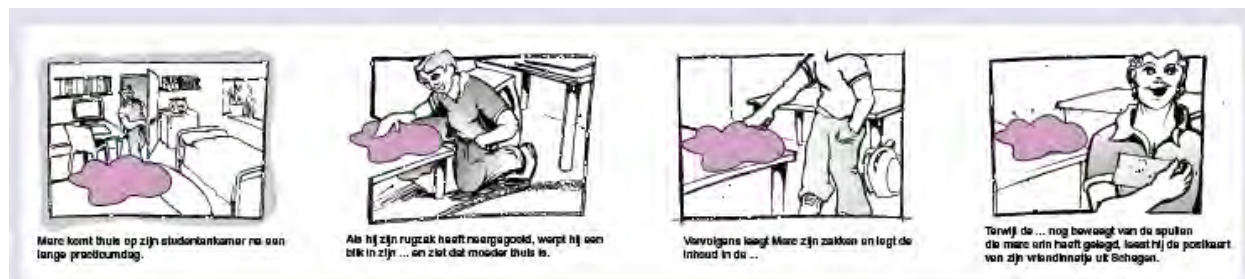


Figure#2.6: This picture of a clock is ambiguous: it can mean different things.

Traditionally in education there has been an emphasis on avoiding ambiguity. For example, the avoidance of ambiguity is stressed in the reporting of scientific research where reproducibility and precision are deemed key values. On the other hand, art, design and creativity thrive on ambiguity. Creating generative tools and applying generative techniques takes skill in working with and comfort in dealing with ambiguity. In fact, it has been established that creative people “tolerate ambiguity well” (Zenasni, Besançon, and Lubart, 2008). So it is perhaps not surprising that designers and design students tend to readily learn and apply generative design thinking. It may be more difficult for people who have been trained in the scientific (i.e., expert-driven) mindset to embrace generative design thinking and to not only tolerate but embrace ambiguity. Ambiguity is a resource to call upon and not a state to be avoided. (Gaver, Beaver, and Benford, 2003). You can use ambiguity as a resource for sparking creativity.

Incompleteness compels people to fill in. Incompleteness is one form of ambiguity. People are very good at filling in holes in an otherwise coherent whole. The act of “filling-in” or “closure” is, in fact, one of the core principles in how we understand comics and graphic novels as Scott McCloud clearly demonstrated in his classic book called *Understanding Comics: The Invisible Art* (McCloud, 1993). By the way, for any reader who is at this point somewhat uncomfortable with these discussions about ambiguity and incompleteness, a reading/looking at *Understanding Comics* is a highly recommended diversion.

Figure#2.7 shows a part of a storyboard about a communication product, in which the product is hidden in each picture by a pink cloud. In this way, the reader is invited to think first of the story, and not yet respond to details of the envisaged product. In fact, a storyboard like this may help the reader to imagine what the product could do or what it might look like.



Figure#2.7: Leaving holes is a storytelling technique that can be used to trigger ideas in people's minds

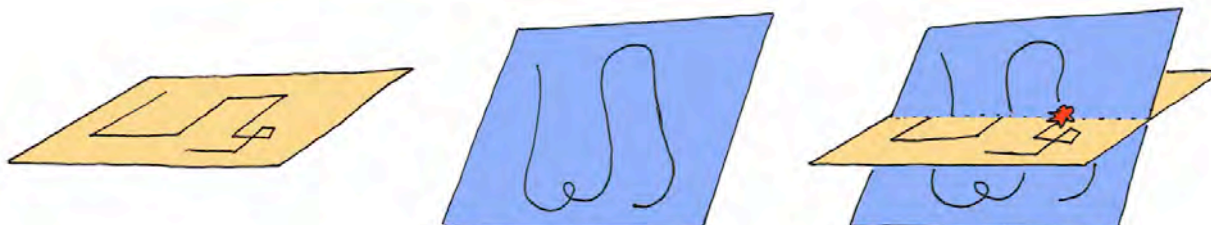
Leaving holes can be used to trigger people's imaginations. Similarly, an open-ended sentence can be a very inviting way to ask a question. Compare the following two ways of asking people to talk about what they do in their free time. By having people complete the sentence, rather than provide an answer to a question, we are likely to learn much more about the person and also get a much wider variety of responses across people.

As a question to answer	As an open-ended sentence to complete
What is your favorite pastime?	When I have nothing to do, I like to
.	
.	

Figure#2.8: Open-ended sentences invite a wide variety of responses.

Creative thinking often works by having people make new connections between previously unconnected ideas. There are two theories about creative thinking that rely on that fact which are helpful to know about: bisociation (Koestler, 1964) and metaphor (Schön, 1963).

Use bisociation to make new connections. Ideas connect to other ideas in two ways: through either association or bisociation. Association links similar or related ideas as we described earlier in the discussion about how ideas can be thought of as clouds of associations. Koestler (1964) introduced the concept of bisociation wherein two apparently unconnected ideas are brought together, and a new connection is explored. *"I have coined the term 'bisociation' in order to make a distinction between the routine skills of thinking on a single 'plane', as it were, and the creative act, which, as I shall try to show, always operates on more than one plane."*



Figure#2.9: Koestler explained bisociation as a new idea emerging from the spark as two concepts (each represented by a line in a plane) are brought together (picture after Koestler, 1964).

Koestler hypothesized that bisociation is the basic creative act and that this act is the same for art, science and humor. Koestler stated that “*every creative act involves bisociation, a process that brings together and combines previously unrelated ideas.*” Koestler’s evidence for bisociation came from a common pattern of intersecting conceptual planes of thought that he has described with many examples showing creative achievements in the arts, sciences and humor.

The principle of bisociation can be used to facilitate idea generation. For example, when brainstorming for extensions of a line of toothbrushes (think of this as the first plane of thought), seemingly unrelated ideas (e.g., swimming) and/or objects (e.g., peanut butter) can be used to set up the second plane of thought. Bisociations in the form of new ideas are very likely to emerge at the intersections of the two planes.

Using metaphor to make new connections. In metaphor, one idea is positioned against another and the fit between the two is explored. According to Schön (1963), this mechanism is the best explanation of how new ideas and solutions are found. He writes:

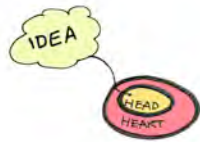
I know what a drum is. I know about snare, bongo, bass, and oil drums. But when I found myself in a metal room with a thin metal wall that reverberated whenever it was jarred, it was a new thought for me that the room was a kind of drum. (...) I am not mistaking a room for a drum (...) There is a change in my concept of ‘this’ (...the room), there is a change in what I have been calling the concept itself (...drum), and these changes are interdependent. (...) These concepts have been displaced to situations outside of their ordinary patterns of use and they have been transformed in the process (pages 30-31)

After the event, both the notion of ‘room’ and ‘drum’ changed in the thinker’s mind. In Schön’s case, from then on, the ideas ‘room’ and ‘drum’ were connected as associations in his mind. In Schön’s view, this is the way a designer, when asked to improve the quality of, say, a ship’s cabin, might use a principle he knew from another domain, e.g., muffling a drum by placing a pillow inside it.

By exploring the connections in the use of metaphor, we learn about both ideas at the same time. In the example of choosing a way to spend next year’s holiday, a mother’s insight that “holiday is like another job to me” calls forth a consideration of comparing how the ideas of holiday and job are similar (e.g., a camping holiday calls for a lot of organizing and planning, preparing dinner under challenging conditions, etc.), but also different (you get paid for a job, but not for going on holiday). A metaphorical comparison is like the collision of two clouds of associations, not like the logical comparison of two dictionary definitions. It explains how two ideas are similar, not whether they are equal.

2.2.2 Creativity and emotion

The second layer in the framework for individual creativity looks beyond the head to the heart, i.e., to one’s emotional state and its role in creative thinking and doing.



Figure#2.10: Emotions play a large role in creative thinking and doing

For many years, cognitive psychologists studied the mind while doing their best to ignore the effect of emotions because emotional states were harder to control and to quantify than were cognitive events. But more recent psychological research has shown that cognition and emotion cannot be separated. In fact, emotion drives cognition, which is shown in the framework by positioning the head within the larger context of the heart. Our emotional states have a large influence not only on how we feel but also on how we think and act, including our ability to be creative. For example, the link between creativity and positive affect (i.e., emotion) was firmly supported in a meta-analysis of 66 studies about creativity and affect (Baas et al. 2008). Alice Isen's (1999) work provides insight into how this works. Her research shows that:

- Positive emotion increases the number of ideas available for association and/or bisociation,
 - Positive emotion increase the breadth of ideas that are relevant to the problem,
 - Positive emotion increases cognitive flexibility, making it more likely that the ideas will connect.
- 1.

Michele and Robert Root-Bernstein (1999) believe that the role of emotion in creativity is fundamental. They write that *“creative thinking in all fields occurs preverbally, before logic or linguistics comes into play, manifesting itself through emotions, intuitions, images and bodily feelings. The resulting ideas can be translated into one or more formal systems of communication such as words, equations, pictures or music or dance only after they are sufficiently developed in their prelogical forms.”*

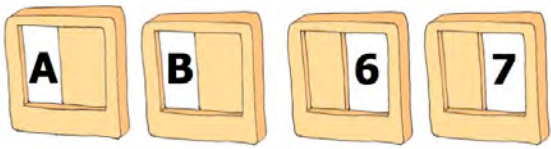
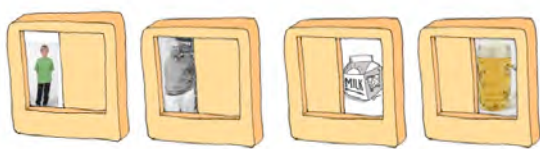
Evoked emotions through the use of ambiguous materials but please be careful. If positive emotional states contribute toward creativity, it is important to facilitate this state of mind in generative sessions where creative thinking and doing is the goal. Giving people a chance to prepare themselves for the sessions helps, as does keeping the atmosphere casual and familiar.

This principle about evoking emotion applies also to the selection of triggers as we will see in the next chapter. But it is not possible to anticipate or to control the kinds of emotions that might emerge in using visual stimuli as triggers in generative design research. If there is any reason to expect extreme emotional responses from people in generative sessions, it is best to be prepared. For example, in a project using generative design research tools and techniques to inform and inspire the design of the new healthcare campus for American War Veterans in New Orleans, we anticipated and prepared for negative emotional responses to photographs because many of the participants suffered from PTSD (Post Traumatic Stress Disorders) as a result of their military experiences. A

clinical psychologist was present for all the generative sessions to manage the negative emotional memories that sometimes occurred.

People think of people and stories best. People are very good at thinking in terms of stories about people and the things they do and what happens to them. This is contrary to the view from higher education where we take great pride in being able to reason in abstract terms, and much of our academic education is directed at training people in abstract reasoning. But most people are not good at dealing with abstractions, as in mathematics and formal logic.

An illustration of this is Wason's (1966) selection task, shown in Figure#2.11. It describes how a simple logical problem, when defined in abstract terms, can be very difficult to solve. But if that same problem is formulated in terms of a story about an everyday situation, we can deal with it more easily.

Abstract rule	Story rule
The left half of each display has a letter, the right side a number. The hypothesis is that if the letter is a vowel, then the right side must be an even number. In which of the four displays must you look at the covered half to see whether the hypothesis is true?	Suppose you are a policeman inspecting whether the 'no alcohol for minors' law is upheld. The left half of each display shows a person in the bar, the right side shows what he or she is drinking. If the person is under 16 years old, then the drink must be non-alcoholic. In which of the four displays must you look at the covered half to see whether the law is abided by?
	 5 year old, 70 year old, milk, beer

Figure#2.11: Solving the task using an abstract rule is more difficult than solving the same task with a story rule.

This finding provides a counterweight to information processing theories which claim that when we encounter problems in the world, we translate them to their basic logical (symbolic) structure, apply logic, and then translate the conclusions back to the real world. Actually, people are not good at that. They're much better at reasoning in terms of story elements.

Our ability to deal with stories lies at the root of the value of narrative techniques in design, such as scenarios, storyboards, and personas. Stories are useful for joining together many different details into a whole. Such story elements provide rich pictures (literally or metaphorically) with which we

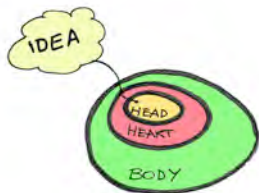
can connect. When designing a new stroller for use in the city, the story about Mary and Jim Jones trying to enter the subway station with their two children aged 1 and 6 is more meaningful to the designer than are ergonomic specifications of the people involved. The ergonomic specifications will become relevant later in the process, however.

Stories are very useful in generative design research because they are engaging and they invoke empathy and imagination. Stories can be true or imagined or even a combination of the two. So storytelling can be a very effective way of triggering ideas about the future.

All senses can be used as generative tools. All senses, but scents exceptionally so, are powerful means of eliciting emotion-laden memories and stories and they can be used as a component in generative toolkits. However, using scents as generative stimuli is best done by design researchers with some experience under their belts because the memories and emotions that come from smells may be so powerful and time-consuming that they derail the session's agenda (Khanna, 2008).

2.2.3 Creativity in motion

The third layer in the framework for individual creativity looks at the whole person in action. So here the head and the heart are in the body moving through time. The implications of this much larger layer for creativity are immense and are just now beginning to be explored.



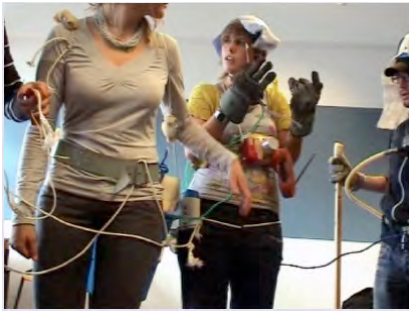
Figure#2.12: Individual creativity takes place in the body and over time

Pretending and enactment can facilitate creativity. Design involves imagining and creating new life situations for people in circumstances that have never been experienced before. Or as Henrik Gedenryd (1998) succinctly stated, “Design is an inquiry into the future situation of use.”

Over the years, designers have developed a range of expressions to help them imagine possible futures, and explore situations in them. These expressions include traditional design visualizations such as sketches, models, and prototypes. But with the larger scale and scope of design challenges we face today has come the need to explore alternative ways to visualize future situations of use. More recently, the complexities around computers, interactive technologies, and mobile services have strengthened the attention for techniques such as enacting, i.e., acting out envisaged interactions with props, rather than merely talking about the interaction. Enacting helps people to

experience the time scales, the complexities, and the effect of body scale. By experiencing new or possible situations in a bodily way, (e.g., Burns et al., 1995; Fulton Suri and Buchenau, 2000; Oulasvirta et al, 2003; Buxton, 2007; Simsarian, 2003; Diaz et al., 2009) things may become apparent which were not evident under abstract consideration alone.

Enacting refers to the use of the body in the environment to express and experience ideas about future use situations. Early in the design process, enactment might take the form of pretending, with future “products” used as props to facilitate the pretending.



Figure#2.13: Enacting future scenarios of use can be facilitated with props

Enactment can also be used later in the design process. For example, if we want to consider the experiential impact of a small-sized toilet space on a train or plane, experiencing it in a full-scale cardboard mockup helps us to form a judgment of it with far more understanding and empathy than we could get from a dimensioned drawing.

Creativity happens over time. It is not an instantaneous event. Graham Wallas introduced what may be the very first model of the creative process in his book *The Art of Thought* in 1926. He identified five stages in the process:

preparation (preparatory work that focuses the individual's mind on the problem),

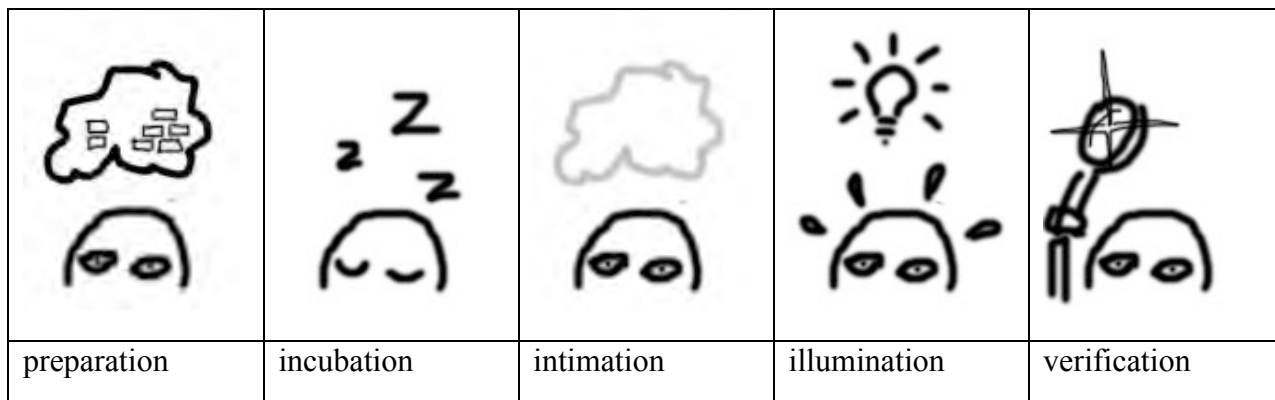
incubation (where the problem is internalized into the unconscious mind and nothing appears externally to be happening),

intimation (the creative person gets a "feeling" that a solution is on its way),

illumination (where the creative idea bursts forth into conscious awareness); and

verification (where the idea is consciously verified, elaborated, and then applied).

Engaging in the creative process (from the perspective of Wallas' 5 stage model) may go something like this: We first immerse ourselves in all the data, then we let go of thinking in an *incubation* period, during which we don't consciously work on the problem, but let it simmer in the background. Often after a good night's sleep, fresh insights turn out to have formed and we try to explore, verify, and elaborate on them. The process of letting go of the problem can be difficult for people who need to feel in control, or who need every step to be rational and proven. To let go of thinking is a skill that typically takes practice before we can gain confidence with it.



Figure#2.14: Five stages in Wallas' model of the creative process

Incubation, a period of downtime, is often an important step in the creative process. It accounts for the observation that when puzzling about a problem, the solution can often hit us unexpectedly after a good night's sleep or in the shower. Famously, Archimedes was said to have come to his famous principle when having a bath, after agonizing about his assignment for a long time.

We cannot expect everyday people to be instantly creative. They will need some time to go through the stages in the creative process and they will need some space for incubation.

Priming and spreading activation are automatic but take time. Priming refers to the phenomenon that you are more likely to remember something if information related to it has been presented previously. Priming is thought to be caused by spreading activation in the network of associations (i.e., the clouds of associations that we discussed earlier in this chapter) in the mind. Priming can occur through associations between words or ideas. For example, a word can prime another word that is 'associated' with it but not necessarily related in meaning. For example, dog primes cat, since the ideas tend to be associated with each other. Pictures can prime and be primed as well.

Priming is automatic and is based on implicit memory events. This means that priming can occur even if the subject is not conscious of the priming stimulus. An example of this was done by Bargh et al. in 1996. Subjects were implicitly primed with words related to the stereotype of elderly people (example: Florida, forgetful, wrinkle). While the words did not explicitly mention speed or slowness, those who were primed with these words walked more slowly upon exiting the testing booth than those who were primed with neutral stimuli.

Understanding how priming and spreading activation work can help us to select and prepare tasks and materials to use in preparing people for creative sessions. Priming and spreading activation may be automatic, but they still take time.

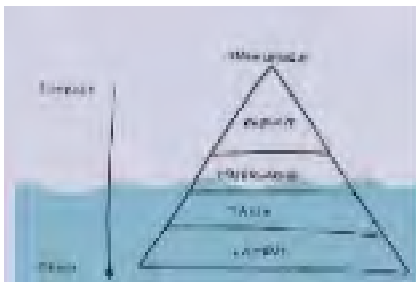
There are four levels of knowledge. Knowledge refers to thoughts and ideas that have already been experienced and have been stored in memory. We can distinguish four levels of knowledge: explicit, observable, tacit, and latent. The examples below relate to the photo in Figure#2.4.

Explicit knowledge can be stated in words, and is relatively easy to share with others; e.g., “Today I have three appointments on my agenda”,

Observable knowledge refers to thoughts and ideas that can be obtained by watching how things happen or how people behave. People are generally not often aware of their own behavior that others can readily observe. For example, “She sometimes makes faces at people when she is on the phone.”

Tacit knowledge refers to things we know but are not able to verbally communicate to others. For example, you probably know how to make a phone call when your hands are full of stuff but this would be difficult to explain to someone else.

Latent knowledge refers to thoughts and ideas that we haven’t experienced yet, but on which we can form an opinion based on past experiences. Latent knowledge will be knowable in the future. It is not easy for people to express this type of knowledge. For example, “I’d like to be able to automatically postpone meetings when I have trouble with my car.”

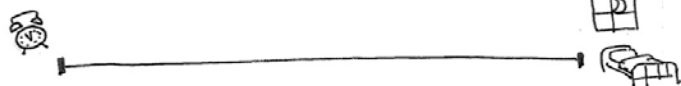
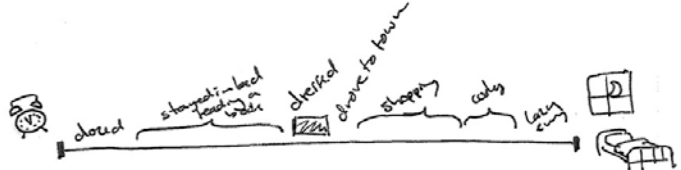


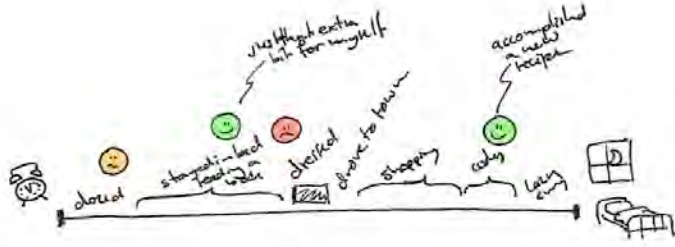
Figure#2.15: Some levels of knowledge are easier to access than others

In Figure#2.15 (after Sanders, 2002) these levels are shown as a pyramid, with two levels above water, and two below. The figure indicates two things: explicit and observable knowledge are only the tip of the iceberg of the things that people know. And it takes effort of looking beneath the water surface, to see what is happening in the tacit and latent levels. Generative design research tools and techniques will give you the means to see what is going on at the tacit and latent levels of what people know.

Layering the levels of knowledge. We want people to reflect on and express their needs and values in order to explore future scenarios of use. But needs and values are abstract qualities that people are not often used to talking about directly. They fall at the level of tacit and/or latent knowledge. If someone asks you “what are your needs and values for travel on a holiday?” it’s difficult to give an answer right away. But given a specific holiday, it is not so difficult to explain why you did or did not like a certain part of it. We can best think about values by linking them to situations or occurrences, that is, to stories.

This observation lies at the basis of generative exercises such as the ‘day in the life’ exercise, which consists of three steps. First, someone is asked to describe the steps that occurred in one specific day. This forms the ‘layer of facts’. When that has been completed, they are asked to explain which of these activities they enjoyed (the high points) and which they hated (the low points), forming a ‘layer of valence’. In the third step, he or she is asked to indicate, for each high and low point along the story, *why* it was high or low. Their explanations about *why* reveal the layer of needs and values. Instead of asking people to attach valence and value to individual facts in isolation, we let them make these judgments within the context of a whole layer. Stepping up from stories about specific days to needs and values is a way of releasing tacit and latent knowledge.

0	the empty timeline	
1	the layer of facts is added	
2	‘smileys’ and ‘frownies’ are added to indicate high and low points	

3	reasons why the points were high or low have now been added	
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Figure#2.16: The day in the life exercise can be used to layer people from stories to a description of their needs and values.

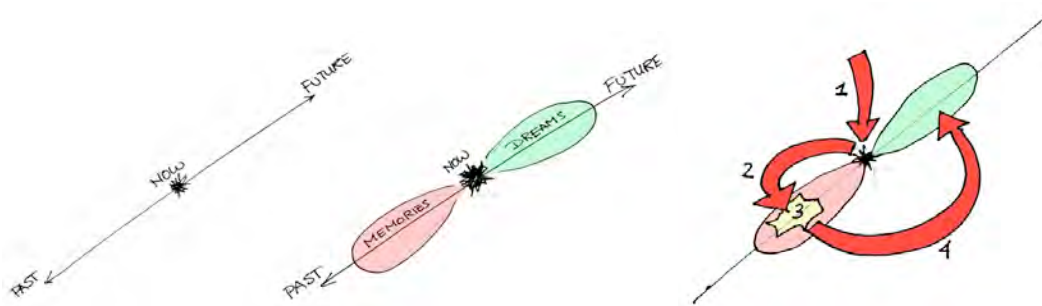
The strength of the layering approach is that it invites people to first create a complete story, then evaluate it, and then find reasons for their evaluations. If we had asked them for a single good or bad event and its reason, it would be difficult for them to answer. But now, when they are asked to evaluate events, they can compare the whole layer that they mapped out in the first step. And when they are asked for the reasons, they can review all the highs and lows that they evaluated before, again comparing the wholes. The day in the life exercise serve two purposes. First, it prepares people for the generative session by getting them to remember and reflect upon their day. Second, it provides the foundation for layering through the levels of knowing.

Sensitizing

Insight into the deeper layers of understanding requires that the participant has been thoroughly involved in the problem or situation for some time. Therefore, people who will be involved in creative sessions will need to be prepared for these sessions before they come. Ideally they will have a week or two for the preparation or immersion process that takes place in a familiar setting such as their home or place of work (see chapter#6, page#X). We often give the participants diaries or workbooks to guide them in daily self-documentation of their thoughts, feelings and ideas about the experience being investigated. Or we may ask them to observe and document their daily lives through photographs or videos. By having them immerse themselves over a longer period, they can become more sensitive to their awakened memories and associations and have the opportunity to gather stories that illustrate things they find interesting or worthwhile. In the generative sessions, we are able to build on this awakened sensitivity and expressive ability.

The path of expression

At the larger time-scale of method, we follow “the path of expression” which guides the time course of the creative process.



Figure#2.17: The experience of the moment is connected to past and future through memories and dreams. The path of expression (right) shows how a person's awareness can be guided in steps by thinking first of the present, then of the past, then looking for underlying layers, in order to move toward the future.

The path of expression is based on a model for understanding experience that is shown in the middle of Figure#2.17. The model illustrates that the concept of experience is focused on the 'moment,' the center of the model marked as 'now'. The moment connects both backward to past experiences and forward to future experiences. Thus, people's memories of past experiences influence how they behave and feel in the moment. Similarly, people's dreams for future experience also influence their perception of the moment. To gain a rich understanding of people's experiences, we give them tools and techniques to explore their current situation as well as past experiences. And in order to invite people to explore future experience, it is vital to provide them with the space and the materials with which to imagine and to make things that they can use to show or to tell their ideas about future scenarios of use.

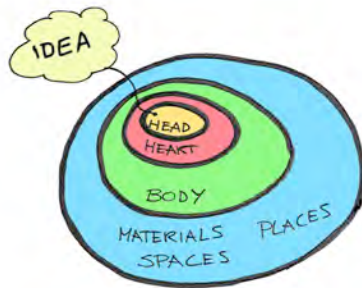
The path of expression is a process that can be used for exploring present, past, and future experience. It serves as a framework for planning the path that the participant will take in a generative design research session, culminating in their ability to imagine and communicate their hopes, dreams and fears for the future. This is how the path of expression method typically works.

1. Participants are asked to observe, reflect on, and describe their current experiences in the immersion period before they come to a workshop session, e.g., through diaries and photo-taking as described above.
2. They are also asked to select, and reflect on, memories of earlier, related experiences. Then during the session, they are invited to share their observations and past memories and discuss current events with each other.
3. This sharing of experiences helps them to access underlying needs and values, which serve as the basis for
4. exploring their aspirations for future experience.

Utilizing the path of expression framework enables people to connect to what is meaningful from their past and present experiences, using that as a springboard for ideation about the future. The path of expression serves as a scaffold upon which to build the journey that the participant will take in a generative design session. Utilizing the path of expression framework also stimulates creativity on the part of our participants. It lets them avoid the pitfalls of becoming fixated on preconceived ideas or their first ideas about the future. In fact, we find that people are often pleasantly surprised and proud of how creative they are in generative design sessions.

2.2.4 Creativity in the environment

The fourth layer in the framework for individual creativity is described by the environment in which the action is taking place. This layer includes the physical environment with its places and spaces as well as all the stuff in the environment such as materials, supplies, props, and tools.



Figure#2.18: Individual creativity is influenced by environmental factors such as places, spaces and stuff

Creativity is fostered by having a choice of spaces in which to explore. The design of spaces that facilitate creativity is a relatively new area of architecture and interior space design practice, but one that is being aggressively pursued today by progressive architecture firms and workplace furniture manufacturers. Some general principles about workplaces that foster creativity and innovation exist. The spaces and places for creativity should:

- have furniture that can easily be rearranged,
- have walls where material (especially visual material) can be posted for all to see and to act upon,
- facilitate a wide variety behaviors such as quiet reflection, relaxation, active collaboration, mess-making, etc.,
- provide both for individuals and for groups of varying size in working face-to-face,
- provide for many moods including playful, stimulating, and informal as well as formal, and
- be open to people both inside and outside of the organization.

It is very important to think ahead about the space in which generative sessions or analysis sessions are to be held. How well do they adhere to these design criteria for creative spaces and places? The contributions to this chapter written by Arnold Wasserman show how these design criteria have been embodied in the form of *charrette spaces* in an innovation consultancy.

The materials present in the environment can also foster creativity. Generative design research has a lot to say about the nature and quality of the tools and other materials that can be used to support creative thinking and doing. The Make toolkits that we will describe in the next chapter are designed to facilitate, support and provoke creative thinking. These toolkits contain a collection of ambiguous (as well as unambiguous) elements, such as words and images for making something such as a collage or a map. These words and images have been selected to trigger associations in the

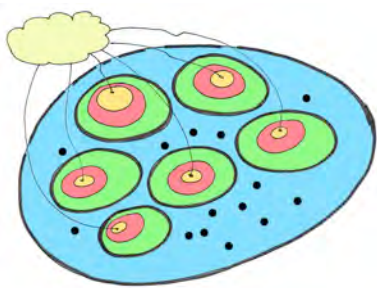
area of experience that is studied. Some triggers are chosen to not be ambiguous so that the participants can start by choosing recognizable stimuli that relate directly to the area of experience. This can help people feel at ease in the creative session. Other triggers in a toolkit are chosen to be ambiguous, so they can evoke a range of associations and can be selected for many reasons. This has two advantages: first, by being ambiguous, it gives the person freedom to interpret the trigger from the perspective of his or her own experience. Second, when that person presents their collage or map to others, the ambiguity of the trigger invites them to explain their interpretation of it and reasons for choosing it.

The physical environment and the materials available there can help to stimulate creative thinking or innovative solutions. We will describe more about the materials for facilitating creativity in Chapter Three.

2.3 Social creativity

“Creativity does not happen inside a person’s head but in the interaction between a person’s thoughts and a socio-cultural context.” (Csikszentmihalyi, 1996)

The framework for individual creativity explodes in scope and scale when we think about people coming together in creative ways. When people come together, the number of ideas and the breadth of the ideas that are brought to the table increases dramatically. This is especially true when people from different backgrounds come together. The diagrams in Figure#2.21 show how we might see social or collective creativity relative to the framework for individual creativity. Note that the artifacts they create help them to connect the ideas of individuals. The artifacts serve as ‘boundary objects’ that serve an interface between different communities of practice (Star and Griesemer, 1989).



Figure#2.19: The framework for individual creativity takes on a whole new life when considered from the perspective of social or collective creativity. The black dots signify materials and artefacts made by the participants.

Psychologists started conducting research on creativity in the 1950’s with a focus on the creative individual because at the time it was thought that the source of creativity and innovation was the individual, the solitary genius. It wasn’t until the 1990’s that this narrow focus was challenged and the role of collaboration in creativity and innovation began to be explored. Keith Sawyer (2006 and 2007) does an excellent job of describing this new perspective on creativity and innovation and the

impact that it will have on schools, organizations and industries in the future. Sawyer claims that all true innovation originates in collaboration rather than through the efforts of the solitary genius, supporting this claim with numerous contemporary examples.

With the increase in collaborative ways of working and with the Internet's ability to connect people, it is clear that social creativity will become an even more important area of research in the future. As was discussed in the Introduction, the only way we will be able to address the challenges of wicked problems and fuzzy pathways are through collective forms of creativity and generative design thinking.

Creative teams are made up of a diversity of people. There is evidence that team creativity is founded on diversity and difference (Nijstad and DeDrue, 2002). In other words, the more diverse the members of the team, the more creative they have the potential to become. Diversity is not always easy because different perspectives often lead to disagreements, but it is a powerful precondition for creativity. Different people bring different ways of thinking and doing to the table. The following sections describe some of the ways in which people differ in thinking, decision making and reasoning.

Balance left and right brain thinking. Brain research in the 70s and 80s pointed out two seemingly complementary sets of thinking skills, which have been associated with left and right brain hemispheres. One set (most often associated with the left brain hemisphere) contains skills in dealing with words, numbers, logic, sequences. These are the skills that have traditionally been pushed forward in academic education. The other set (most often associated with the right brain hemisphere) contains skills in dealing with rhythm, color, Gestalt, graphical composition, and visualization. These skills have received attention in arts curricula. Recent wisdom supports the fact that both sets of thinking skills are important, and that in facilitating people to use their minds, we should attend to the needs and powers of both 'channels'. (e.g., Pink, 2005).

Utilize Henry Mintzberg's three strategies for decision making. Different types of problems call for different strategies. Mintzberg sketches three different strategies for decision-making that can be used in different situations: 'thinking first', 'seeing first' and 'doing first'.

Approach	Thinking first	Seeing first	Doing first
Features the qualities of..	science	art	craft
	planning, programming	visioning, imagining	making, trying it out
	the verbal	the visual	the visceral
	facts	ideas	experiences
Works best when..	• The issue is clear • The data are reliable - The context is structured	• Many elements have to be combined into creative solutions • Commitment to	• The situation is novel and confusing • Complicated specifications would

	• Thoughts can be pinned down • Discipline can be applied	those solutions is key • Communication across boundaries is essential	get in the way • A few simple relationship rules can help people move forward
As in..	an established production process	new product development	companies that face a disruptive technology

Figure#2.20: A comparison of three strategies for decision making from Mintzberg and Westley (2001).

Thinking first works well for situations where we understand the situation well, and have good models such as in a high school science assignment, where the teacher knows the right answer. But it doesn't work well in cases where we know 'too much or too little', i.e., there is lots of missing, uncertain or conflicting information.

In the latter case, seeing first (e.g., sketching a vision or using the approach typically trained in art) works better. Seeing first works well in creating future scenarios on the basis of incomplete data. Finally doing first (creating a prototype, and trying it out, and improving it iteratively) works best if we have very little experience with the situation at hand: i.e., we know very little. For example, if you want to learn about building sand castles that can withstand a wave, build one on the beach and notice how and where it goes wrong. Mintzberg explains that people tend to have a preferred strategy for decision making and that they may benefit from being exposed to and being able to practice the other strategies.

From the perspective of social creativity, it is ideal to have a team made up of people who together have all three strategies for decision making as their preferred mode.

Understand Charles Peirce's three modes of reasoning. Charles Sanders Peirce, an American philosopher, logician, mathematician, and scientist, introduced three modes of reasoning that provide another window into the workings of the mind in action: deduction, induction, and abduction. The table below gives examples of each mode of reasoning.

	Deductive reasoning	Inductive reasoning	Abductive reasoning
Before	Shops sell goods only from 9 to 5 & This cake was bought in a shop	I looked at many ways of making cakes & Always the cake was made with flour	A birthday cake should be sweet; & Honey can make things sweet
After	Certainly, the cake was bought between 9 and 5	Probably flour is a necessary ingredient for making a cake	Perhaps this birthday cake could be made with honey

Figure#2.21: A comparison of the three modes of reasoning

Deductive reasoning starts from a general rule such as ‘people always do something in this way’, combines it with a particular statement ‘this person does this something’ and ends up with a particular conclusion such as ‘then this person must have done it this way’. Deductive reasoning is a purely logical thing. If you accept that the two statements are true, then the conclusion must necessarily be true. It requires no looking at the world. Of course, if the starting rule is wrong (e.g., there are shops whose opening hours exceed 9-5) the conclusion may not be correct (or it may still be correct, but not because of our reasoning).

Inductive reasoning happens when researchers try to establish general rules based on observation in the world. This is how the experimental sciences work. One typically cannot, e.g., see all the possible cakes that exist. So research methodologies have been developed over the past 200 years, often involving statistics, to guarantee that ‘looking many times’ can be sufficient to show that the conclusion can be trusted with a sufficient degree of confidence.

Finally, abductive reasoning is closely related to creativity and design. It is radically different from the other two, because it is about inventing a possible solution for a problem. If you want to explain how a birthday cake can be made to taste sweet, honey can be one explanation but other explanations (sugar, ginger syrup) are also possible. Abductive reasoning is finding an explanation, not proving it is the only one. Abductive reasoning is a form of inference or a leap to a conclusion. Abductive reasoning has recently gained a lot of attention as a core element in design thinking. Unlike deduction and induction, where the conclusion follows from the inputs, creativity is an abductive process of finding a new and possibly fitting solution from ‘somewhere’.

In summary, the primary advantage of social or collective creativity is that people with many different ways of thinking, decision making and reasoning can be brought together, increasing the chances that connections and new insights will occur at many levels. But at the same time, with all these people having different ways of seeing, thinking and doing, it may be hard for them to come to agreement. The generative design tools and techniques can help here too. They support them in developing a common language, a transdisciplinary design language that everyone can use to express their ideas.

2.4 Moving on

In this chapter we have presented ideas about thinking and creativity in order to provide a foundational layer for understanding how and why generative tools and techniques work. We reviewed key principles and theories about thinking creatively, first from the perspective of individual creativity and then introduced principles and findings that describe social or collective creativity. As a generative design researcher, you need to understand both perspectives because sometimes we ask people to work alone and at other times we ask them to work together.

We discussed that we cannot ask people to become instantly creative. They need time for immersion and for spreading activation to take place in order to build from the past and present and into the future. The path of expression is a means by which the journey to the creative expression of meaningful ideas can be scaffolded.

How can we invite ordinary people, whose creativity is probably latent, to join the design development process? First, we must learn to respect their creativity. Then, we need to provide them with experiences and tools so that they can exercise their creativity and participate directly in the design process. The next chapter gets very much more concrete about putting all this thinking about creativity into action.

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CHAPTER 3 How it works: Generative tools and techniques

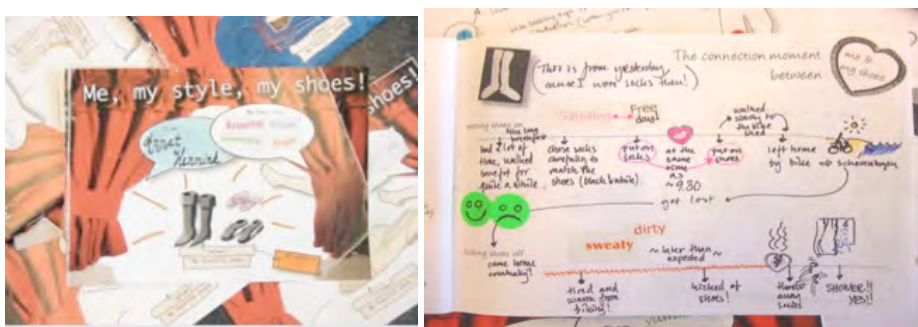
Overview of Chapter Three

In the previous chapter we talked about how people think creatively. In this chapter we describe how to put the principles and theories of creative thinking into action. We introduce a wide range of tools and techniques that can be used in generative design research, organized according to a simple, people-centered framework: What people say, what people do and what people make. Tools and techniques for ‘what people say’ and ‘what people do’ have been practiced for many years, so we will not attempt to describe those in detail. To do so would take us beyond the scope of this book. Instead, we will focus on the category of tools and techniques that fall in the ‘what people make’ category which is unique to this book. We will also discuss the importance of the interplay of all three categories. This chapter serves as the springboard for Part Three where the details of planning, executing, documenting and analyzing generative design research are covered.

An infinite set of tools and techniques

Over the past decades a wide variety of techniques have been employed in learning about people, and in learning from people about their everyday experiences. These techniques have come from a variety of practices, both in industry (marketing) and in academia (psychology, anthropology and sociology). Some techniques and tools have crossed over between practices, and have been refined and enhanced in the process.

First, let’s make clear the difference between the terms ‘tool’ and ‘technique’. By *tool* we refer to a physical thing that is used as a means to an end. By *technique* we refer to the way in which this tool is employed. Pencils, pens, and markers are tools for the techniques of sketching, drawing, or annotating. With a description of a tool, we tend to focus on its form; with a technique, on the way it is used.



Figure#3.1 Parts of a workbook that was used to explore the relationship between people and their shoes.

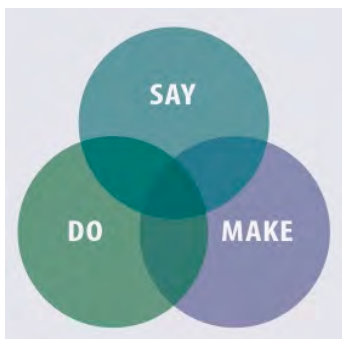
A workbook (Figure#3.1) is a tool, typically in the form of a small book with exercises and assignments, containing text and images. The workbook can be used in different techniques, e.g., it

can be sent to people who carry out the exercises and return the results by mail without the researcher having direct contact with them. In another technique, filling out the workbook is seen primarily as a way of preparing the participants for later activities. So even though the tools may have the same physical form, the manner in which they are used can differ. Such differences can have large consequences for the insights that are gained and for the way the design process gets informed or shaped.

Say, Do, and Make tools and techniques

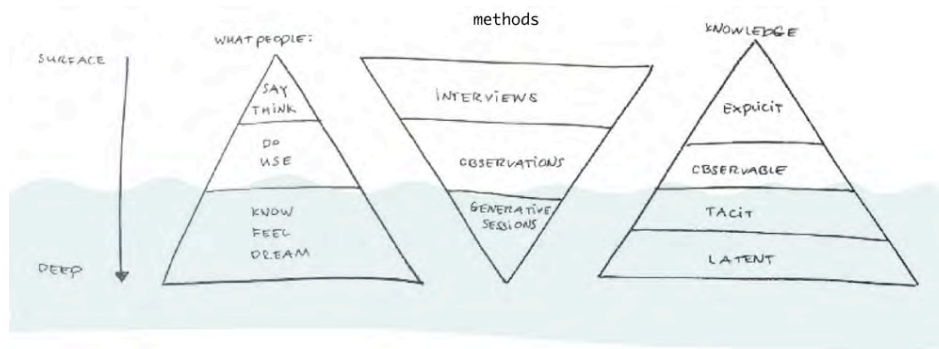
There are dozens of techniques and hundreds of tools to work with the techniques. Techniques can be classified by different criteria, e.g., the way they depend on a theory about the topic being investigated, the type of data that they deliver, or the level of academic rigor, training or financial investment they require. In our experience, the most helpful perspective from which to organize tools and techniques is people-centered, focusing on the activities of the participants rather than those of the researcher or the data. One can look at what people do, what they say, and what they make. Many studies (particularly in market research applications) include only Say techniques, but it is becoming increasingly common to find combinations of Say and Do techniques being used.

As you might expect from reading Chapter Two, it is the latter category, what people make, on which we lay the most emphasis in this book. That is in part because it's the newest in the context of research techniques, as it borrows from design. It's also because the Make category provides more opportunities for exploring experience at a deep level. But as we will explain later, Make tools and techniques should not be used in isolation. A generative study almost always has elements of all three: Say, Do, and Make.



Figure#3.2 Say, Do, and Make tools and techniques complement and reinforce each other.

For example, if you're conducting a generative study on future kitchen experiences of people, you can visit their homes and observe what they do: how do they use the kitchen? You can ask them questions and listen to what they say: interview them about what they do in the kitchen, with how many people, for how long, and when. You can get them to recall earlier kitchen experiences and reflect on those. And you can study what they make when given an 'ideal kitchen experience construction kit'; what ideas do they have, and what reasons do they give for these. The path of expression (see chapter#2, page#X) guides the way in which you weave together these inputs from people to construct valuable concepts about possible futures.



Figure#3.3 Methods that study what people Say, Do, and Make help access different levels of knowledge.

All research techniques in use today for exploring people's experiences fall into one of three categories – what people say, do or make – or they fall into the areas of overlap between the categories.

What people do

With Do techniques, somebody observes people, their activities, the objects they use, and the places where they conduct these activities. This somebody doing the observing can be the researcher, or it can be the people themselves. Self-observation and self-report may, in fact, be the only option available for certain everyday activities such as those involving personal hygiene, since this is a domain that people prefer to keep private. Of all three categories, the Do category seems closest to scientific practice, because it can be conducted by an unobtrusive researcher 'objectively' observing and recording the behavior of people. In practice, though, there are limits to the level of unobtrusiveness that can be attained, e.g., for legal reasons of privacy, and for practical reasons. A great number of tools can be used for making observations of people and recording traces of use. For example, photo- and video-cameras, note-taking paper for writing and sketching, tally sheets, voice recorders, etc. Viewing and documenting the place of the action itself, even without people in it, is also of great value. Gosling et al, (2002; cited in Gladwell, 2005) showed that looking around someone's living room for fifteen minutes can give you a more reliable impression of that person's character than spending a day with the person him- or herself. There are a vast number of options available for studying what people do. Three salient dimensions to consider are the observer, the level of intrusion, and the recording media used in the study:

- **Who does the observation?** The answer can vary from researcher-as-observer to participant-as-observer with the points in between describing hybrid conditions along this scale.
- **How intrusive is the research?** On the one end, the observations are so low-key that the people being observed don't notice (e.g., with hidden cameras). On the other end, the observations may be very visible, e.g., when the researcher accompanies the participant as a photojournalist. The obtrusiveness of the observer may depend on where the research takes place (e.g., at people's homes, in public spaces), and whether it is a planned activity or not. If you are documenting the

observation of people, informed consent is a legal requirement (*see contribution on page#InformedConsent*).

- **What media are used** to document the phenomenon? Again, there are many options and various media through which one can document behavior and traces of human activity: eyes, words on paper, diagrams on paper, photography, video, etc. The new communication technologies such as picture phones with video-recording and text messaging are opening up the possibilities along this dimension.

Two useful resource for learning more about the Do techniques include *Doing Anthropology in Consumer Research* (2007) by Patricia L. Sunderland and Rita M. Denny and *Ethnography: A Way of Seeing* (2008) by Harry F. Wolcott.

What people say

It's hard to imagine a reader of this book who has not been a respondent in a Say technique. Questionnaires, polls, and interviews are different ways of getting answers from people, usually by asking them questions. The form of Say technique that you've probably witnessed most often is the closed form of the questionnaire, in which all questions have been unambiguously formulated, and you are provided with constrained opportunities to answer. For example, 'do you own a bike? (yes/no)', 'what's your gender? (male/female)', 'your age?' (a number range between 20-80)'. The advantage of this form of questioning is that the data can be obtained without the researcher's being present and it can be processed very efficiently. The data can also be subjected to quantitative statistics (e.g., 43% of the respondents were male and owned a bicycle; on average the female bicycle owners were 3.4 years younger than the male ones; etc.).

Say techniques come in gradations of 'objectiveness'. Questions can also be framed more subjectively, e.g., 'how often do you travel by bike? (very often, frequently, rarely, almost never, absolutely never)'; 'indicate on a scale how comfortable your current bike is (1-7)'. Here also, descriptive statistics can be used to summarize the data. Another difference in the range of Say techniques is the amount of openness permitted in the answer. In the examples just described, the respondent could only fill in a limited set of answers to the questions. The researcher had prepared not only the questions, but also the answers he or she expected to get, with very little possibility for initiative on the part of the respondent. However, all of us have experienced questionnaires where none of the answers were appropriate and we've wanted to fill in something completely different but were not given the opportunity to do so. The closed-form questionnaire is one extreme within the 'say' spectrum.

Interviews are Say techniques that allow more freedom to learn from people, and for them to shape the direction that the questions and answers go. Interviews can also be closed in form, but are usually structured more loosely. For example, the interviewer can come prepared with a list of questions, or with a list of issues that serve as a starting point from which the conversation develops in a free form. The value of the loose structure is that the researcher can learn in unexpected directions. However, the price is that he or she has to cope with unexpected responses that cannot be classified and organized right away. And the researcher must be aware of what the boundaries of the question are in order to sufficiently steer the conversation in order to end up with usable insights.

The Say techniques go beyond the superficial layer of behavior; participants can express opinions, voice needs, indicate reasons, and report on events that occurred before the interaction with the researcher. Also, the Say techniques feature not only acts of behavior of a person on his own, but acts of communication of a person toward another. The receiver colors the message, e.g., by his or her own interpretation. And the sender colors the message, e.g., by answering what he/she thinks that the receiver wants to hear. This results in the say/do dilemma: what people say is different from what they do. In other words, people may not practice what they preach, and the views they express may be biased by a basic inclination to please the interviewer. Or they may express views to make themselves look better than they actually are. And worse yet, they might express views to deliberately sabotage the research findings.

There are several dimensions along which to take a position when considering Say techniques. Three prominent ones include the people who do the talking, the amount of pre-determined structure, and again the media/form that are used for the ‘conversation’.

- **Who talks?** Interviews can be one-on-one, but there are many variations, both regarding the number of interviewers and the number of interviewees. A pair of interviewers can interview a single participant, a single interviewer can facilitate a group of participants, and sometimes participants are given tools with which to interview each other.
- Is there a **pre-determined structure**? Some interviews are no more than interactive questionnaires, with the interviewer guiding the interviewee through a fixed set of questions. A less fixed form can be an interview where the interviewer has a list of focus points ready, or even an open-ended interview with very little demarcations beforehand.
- The **media/form dimension** addresses the form that is used to carry on the conversation such as interviews, group sessions or surveys that can be conducted face-to-face, online, through the mail or on the telephone. The media dimension also describes how the Say techniques are documented including audio-recording, video-recording, or note-taking, etc.

A useful resource for learning more about the Say techniques (that also has a brief introduction to the Do techniques) is *Real World Research: A Resource for Social Scientists and Practitioner-Researchers* by Colin Robson (1993).

What people make

Finally, you can have people make things to express their thoughts and feelings. The Make tools and techniques borrow from design and psychology, and involve participants by having them perform a creative act with respect to the subject under study.

An important part of generative techniques are toolkits for expression. These toolkits are carefully developed by the research team to support the participants in a pre-determined activity such as recalling memories, making interpretations and connections, seeing and explaining feelings, or imagining future experiences. Creating a toolkit that is fit for the study is a key skill and key factor to success. Equally important as the physical toolkit is the instruction that is given to the participants, and the way the participants are supported by the facilitator. In the remainder of this chapter we'll give some general principles and guidelines on how the toolkits are constructed. The external contributions throughout this book also show many different examples of toolkits used in practice. They also emphasize the fact that there is not ‘a single toolkit’ that can be used everywhere

and that constructing the toolkit is part of the generative design research process. Chapter#6 describes how the toolkits are put to use.

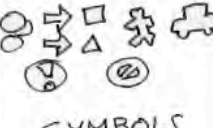
The generative techniques discussed in this book rely on the creative process and usually result in the making of an artifact. In creating artifacts, in engaging in an act of design, we are forced to take into account competing ideas and to resolve ambiguities to make a good enough single, embodied, solution. This is a powerful way of reasoning, which forces confronting all ingredients in a problem, choosing (even if only temporarily) an idea for a solution, and making explicit statements on all its ingredients. It prevents us from ‘hiding in abstractions’, and forces us to commit to an idea. In this making process, important insights arise, which can be made explicit in the presentations that generally follow the creative process.

As to the dimensions of Make tools and techniques, again there’s plenty of choice. For instance, dimensionality, content, and time.

- **Dimensionality:** toolkits can help people to create artifacts with two-dimensional or three-dimensional materials. People can also create static artifacts or scenarios that unfold over time.
- Topic **content** can be cognitive and functional on one end to emotional and expressive at the other end.
- **Time** can be an important choice to make: do you focus on mapping how events might unfold over time or do you focus on a single point in time?

Ingredients of Make toolkits

There is an infinite set of toolkits, and a great variety of types of ingredients. The toolkit that’s most often shown is the word+photo collage-making toolkit. Figure#3.4 shows a set of examples of trigger sets which have been used often in toolkits. Trigger sets are sets of elements designed to trigger associations and/or memories. Elements can be 2D or 3D, words, photos or a variety of other pictorial forms, they can be abstract geometrical forms or representational manikins; each of these types has its own advantages and limitations.

 PHOTOS	Photos tend to elicit emotions and memories, suggest complete situations and stories, and carry many different layers of meaning and associations.	 PUPPETS	Puppets can be used to provoke storytelling and to set the stage for exercises in empathy.
 WORDS	Words are powerful at expressing abstractions such as symbolic meaning or emotional content. Words are also good starting triggers for people who are more accustomed to using words vs. thinking with pictures.	 VELCRO SHAPES	Velcro-covered 3D shapes can be quickly assembled into rough product 'prototypes' and smaller add-on components can be used to express functionality.
 SYMBOLS	Symbolic shapes support making abstractions and formulating general relations, patterns, and rules.	 TINKERING STUFF	Raw collections of scrap materials can be used for constructing objects or for embellishing rough prototypes.
 CARTOON PICS	Cartoonlike expressions often leave room for a variety of interpretations. They can also add an element of fun.	 CONSTRUCTION KITS	Lego and other construction kits are also useful for prototyping concepts.
 SYSTEMATIC DIMENSION SETS	Systematic sets can be used to suggest and express values across an entire dimension, such as a systematic collection of emotional expressions, or a set of body postures.		

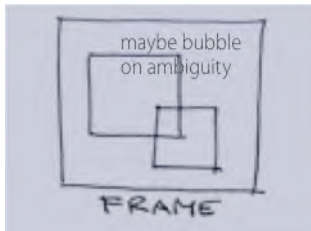
Figure#3.4 Some ingredients that can be used in Make toolkits and where we have observed them to work well

Trigger sets are not generic. They are created for a specific study. The form of triggers suggests certain ways of using them, but the openness leaves great freedom to the participant to use them in expressing his or her intended meanings. The participant can choose whether or not to use them in the Make exercise.

The composition of the set is important. The trigger set should be sufficiently:

- **varied in content.** For instance, a photo collection can contain pictures of places, things and people in certain states, moods, activities, or roles.
- **varied in abstraction**, e.g., including both abstract phrases like 'staying in touch' and concrete things like a picture of a specific mobile phone.
- **varied in levels of ambiguity and openness.** A trigger is rarely characterized by a single meaning. It may suggest different meanings or ways of using it; the participant can decide which to use. For instance, a photograph of a running person can show gender (male), role (policeman), activity (running), mood (tense), feeling (stress), dress (uniform), function (authority), and can be picked for any of these reasons by the participant. Similarly, we've seen the blue rectangular frame in Figure#3.5 used to stand for 'a house', 'a painting', 'a frame', 'a jacuzzi', 'a black hole', 'a cubicle', ...

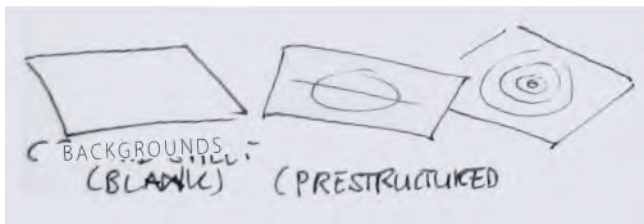
- **varied in aesthetics.** Participants should feel free to express themselves in their own style. Having a trigger set with a variety of aesthetics encourages them to do so.
- **varied in form.** For instance, words in a word set can be presented in randomized order, colors, and fonts.



Figure#3.5 An ambiguous figure such as this frame contains many meanings; a very specific picture will only support very few meanings.

Backgrounds and backdrops for Make toolkits

A backdrop or ground plane is usually provided for the making activity. The backdrop may be blank or it may carry a suggestive structure to guide or to focus the participant's efforts (See Figure#3.6). Whether the participant actually uses the structure provided (or prefers to work on the blank reverse side) can often be left to the participant's discretion.



Figure#3.6 Various backgrounds can be used to suggest or evoke structures to help people get started and to help focus the creative process.

Types of Make toolkits

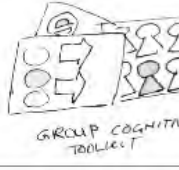
Trigger sets, together with a background and supplies such as scissors, colored markers and glue (or tape) form a toolkit, which is administered according to specific instructions. As we said, the ingredients of the toolkit are important but the way you introduce it to people is equally important. Instructions for administering Make toolkits will be covered in Part Three of this book.

Toolkits themselves are varied in composition, and participants vary greatly in how they will (want to) use the toolkits. For example, an emotional collage toolkit consists of both photographs and words. Many people will use both, but an occasional person will only use words; yet others prefer to write words themselves, or make small sketches. Figure#3.7 shows some types of toolkits.

There are many factors to be considered in creating a toolkit, such as:

- **time and budget** available (both from the time needed by the participant in making the expression, and the time needed by the researchers in creating the toolkit)
- **location of use** (the lab, participants' homes, or public spaces lead to different opportunities, e.g., in available surface area, or freedom of movement)
- **topic of the study** (whether you are looking for concrete descriptions of work practice, or emotional recollections of past experiences)
- **comfort level** (some participants may be at ease in expressing something about the topic, others may be more restricted)
- **what you intend to do** with the findings (e.g., will this drive design of a product, service, organization, communication strategy?)

The size of the toolkit is also very important. It should not be overwhelming, burdening the participant with too many things to consider. For instance, a word-and-photo set which is meant to be used in 10 to 15 minutes collage-making, typically consists of about 100 words and 100 images.

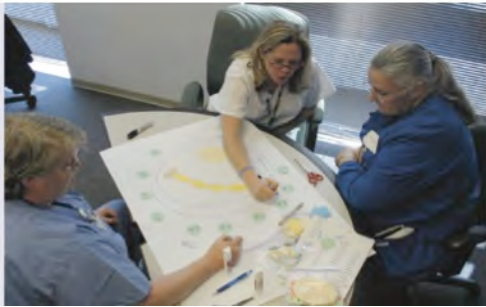
 EMOTION TOOLKIT (SOLO)	Photos and words (or phrases) are used for eliciting memories from the past. The backdrop might be blank, allowing the participant to define its structure. Or the backdrop may be predefined to elicit specific content. For example, good memories inside the circle and bad memories outside the circle.	 GROUP COGNITIVE TOOLKIT	When a toolkit is created for group use, its elements must be bigger, so that it can be handled by multiple people at the same time, and be read from a distance.
 COGNITIVE TOOLKIT (SOLO)	A range of simple and symbolic shapes combined with words can be used to express relationships between ideas or components. For example, how does teamwork work?	 DOLL'S HOUSE KIT	For focused applications, especially later in a session when attention is drawn to creating concrete solutions, toolkits can be more representational instead of abstract. They can also be scale models or 1:1.
 STORYLINE KIT	This toolkit is optimized for the expression of a story. A timeline running from left to right defines the flow of time. Visual and verbal elements might be included to facilitate thinking and expression.		

Figure#3.7 Some types of complete toolkits we have used many times

Applying the Make toolkits

Make toolkits can be used in different settings: by people on their own in preparation for a meeting, in a group, or in a 1:1 interview setting. There are an infinite number of ways to use the toolkits and many different techniques to facilitate their use.

Figure#3.8 and Figure#3.9 illustrate some of these techniques that come from our most recent personal experiences. There are many other examples of Make techniques in use from other contributors throughout the book.



Figure#3.8 Nurses co-creating a concept for ideal workflow on a patient floor. Note that the toolkit components are round, helping them to think in terms of zones and activities, not rooms. This session preceded the one shown below. (NBBJ/rev with Moffitt Cancer Center, 2006).



Figure#3.9 Nurses co-designing the ideal future patient room using a three-dimensional toolkit for generative prototyping. (NBBJ/rev with Moffitt Cancer Center, 2006)

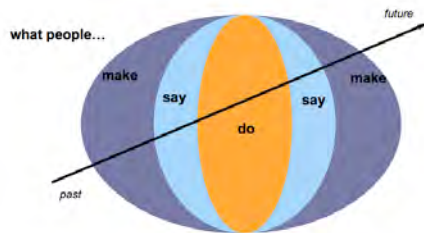
Say, do, make and the path of expression

Depending on the topic of study, the time and the budget, all three types of research tools/techniques (Say, Do and Make) are likely to be used to different degrees. Combinations of the types can provide extra value because one finding may corroborate the other. For example, a Make activity is generally followed by a Say activity: people make a collage expressing their feelings about a topic, then describe in words the reasons behind the choices they made in creating the collage. And a session may begin with an emotion toolkit (to evoke memories), followed by a cognitive toolkit (to explore underlying motivations), and the doll's house kit (to create desired future image).

And often, techniques have ingredients of multiple types. For instance, the researcher can observe interactions between participants when a group of people makes a collage together (e.g., Postma & Stappers, 2006). Contextual inquiries contain interviews with people, conducted in their workplace environments. Such inquiries focus on activities in which the respondents perform actions and discuss these with the interviewer. Another example is in moblogging, for example, sending people an sms/texting on their mobile asking 'what are you doing now?' This technique contains both do and say, because people will tell you what they are doing.

In Chapter Two, the path of expression was explained as a strategy for taking the participant from observing the present, then back to their memories of earlier experiences, then forward to dreams for

desired futures (and possibly to nightmares of feared futures). The three categories of techniques play different roles in the path of expression, as shown in Figure#3.10.

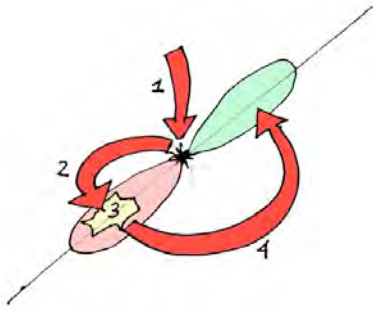


Figure#3.10 The three categories of research techniques address different portions of the timeline of experience. That is why it is important to use techniques from all three categories

Do techniques are about observing the present. They can be very factual and precise. But they are also tied down to this present: they contain no ‘what if’s...’. Say techniques ask for expressed opinions and interpretations, and therefore allow a wider scope about experience that is beyond the present. But Say techniques are limited to that knowledge which participants can readily recall and which they can put into words. The Make tools reveal deeper levels of understanding and can access both tacit and latent knowledge. The Make tools are essential for encouraging people to engage in associative, bisociative and creative thinking. For these reasons, the Make tools are better able to support the jump to imagining the future on the basis of deeper interpretations of the past.

In a generative session, the jump to the future may be facilitated by having the participants take part in the following series of activities, following the path of expression, as shown in Figure#3.11 (see also Figure#2.18 on page#tbd):

- 1 start with observing and documenting their *current* activities (what people do) around the topic of the study
- 2 then recall memories from *earlier* experience using a make exercise that includes photographs and other evocative triggers
- 3 then express ideas and possibilities for the *future* with a make exercise that allows for abstract and/or experiential expression
- 4 and finally engage in a make exercise with a more concrete Make tool such as *Velcro*-modeling to create artifacts for future experiences.



Figure#3.11 There are four steps in the path of expression across the timeline of experience

In this chapter we described the tools and techniques, and their underlying principles. The external contributions throughout the book provide a view on the variety of shape, complexity, and aesthetics that these can take. As mentioned before, the tools are collaborative, they are expressive of the topic of experience, and of the different people involved. In Part Three, especially in Chapter 6, we go into detail on how to use them in practice.

References

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